More Efficient Energy Management for Hybrid AC/DC Microgrids With Hard Constrained Neural Network-Based Conversion Loss Surrogate Models

Xuan Liu, *Graduate Student Member, IEEE*, Yuanze Wang, Xiong Liu, *Senior Member, IEEE*, Tianyang Zhao, *Senior Member, IEEE*, and Peng Wang, *Fellow, IEEE*

***Abstract*—Conversion loss modeling plays a crucial role in hybrid AC/DC microgrid (MG) energy management (EM). However, accurate calculation of the conversion losses is often very costly. Additionally, existing surrogate models typically rely on fixed-voltage DC buses, leading to excessive voltage magnitudes. To overcome these limitations, we propose surrogate models based on piecewise linear neural networks (NNs) that estimate conversion losses using converter power and variable-voltage DC buses. These NNs employs the Rectified Linear Unit (ReLU) activation function, allowing reformulation into linear constraints. During NN training, additional hard constraints are enforced via the augmented Lagrangian method to enhance feasibility. To limit the computational cost of the reformulated NN constraints, weights sparsity and ReLU stability regularizations are applied to the NN training process. The efficient energy management is implemented as a two-stage stochastic unit commitment (UC) problem with integer recourse. To solve this UC problem, we propose a finite iteration convergent Benders decomposition algorithm with several acceleration techniques, which reduce the computational burden significantly. Our case study demonstrates that: 1) the proposed surrogate model accurately approximates conversion losses while satisfying hard constraints, 2) optimizing the DC bus voltage leads to more efficient system operation, energy savings, and better overall performance, and 3) combining the NN weight optimization with Benders decomposition acceleration techniques significantly reduces the computational associated with the NN surrogates.**

***Index Terms*—Hybrid AC/DC microgrids, power conversion losses, linear surrogate models, variable DC bus voltage control, neural networks, mixed-integer linear programming.**

Manuscript received 31 October 2022; revised 19 March 2023, 19 July 2023, 30 September 2023, and 28 November 2023; accepted 9 December 2023. Date of publication 12 December 2023; date of current version 23 April 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 52277182, and in part by the Major Talent Program of Guangdong Province under Grant 2019QN01L109. Paper no. TSG-01630-2022. *(Corresponding authors: Xiong Liu; Tianyang Zhao.)*

Xuan Liu, Yuanze Wang, Xiong Liu, and Tianyang Zhao are with the Energy and Electricity Research Center, Jinan University, Zhuhai 510630, Guangdong, China (e-mail: xliu514@gmail.com; wangyuanze@stu2020.jnu.edu.cn; liushawn123@ieee.org; matrixeigs@gmail.com).

Peng Wang is with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: epwang@ntu.edu.sg).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TSG.2023.3342101.

Digital Object Identifier 10.1109/TSG.2023.3342101

NOMENCLATURE

Indexes and Sets

$g \in \mathcal{G}$	Generator set
$d \in \mathcal{D}$	Demand set
$c \in \mathcal{C}$	Bi-directional AC/DC converter set
$v \in \mathcal{V}$	Bi-directional DC/DC converter set
$t \in \mathcal{T}$	Scheduling time period set

Parameters

$c_{\mathrm{U},g}, c_{\mathrm{D},g}$	Start-up and shut-down cost of generator g [\$]
c_{L}	Cost of power conversion losses [\$]
c_{S}	Cost of load shedding [\$]
$\underline{\mathrm{UT}}_g, \underline{\mathrm{DT}}_g$	Minimum up and minimum down time of generator g [h]
$\overline{\mathrm{UT}}_g$	Maximum up time of generator g [h]
R_g^+, R_g^-	Ramp up and ramp down limitation of generator g [kW/h]
$\underline{P}_g, \overline{P}_g$	Minimum and maximum power output of generator g [kW]
$\overline{P}_{\mathrm{UG}}$	Maximum power exchange between MG and the utility grid [kW]
$\overline{P}_c$	Maximum power of converter c [kW]
$\overline{P}^t_{\mathrm{PV}}$	Maximum PV output at time t [kW]
$\overline{P}^{\mathrm{ch}}_{\mathrm{ESS},v}, \overline{P}^{\mathrm{dc}}_{\mathrm{ESS},v}$	Maximum charging and discharging capacity of battery (connected to converter v) [kW]
$\underline{E}_{\mathrm{ESS},v}, \overline{E}_{\mathrm{ESS},v}$	Minimum and maximum energy limit of battery (connected to converter v) [kW]
$p_{\mathrm{D}}^{\mathrm{AC},t}, p_{\mathrm{D}}^{\mathrm{DC},t}$	AC and DC loads at time t [kW]
Δt	Time step [h]

Binary Variables

α_g^t, β_g^t	Start-up and shut-down command of generator g
u_g^t	1 if generator g is online at time t, 0 otherwise
$I^t_{\mathrm{ESS},v}$	1 for battery (connected to converter v) charging at time t, 0 otherwise
I_c^t	1 for converter c rectifying at time t, 0 otherwise

Continuous Variables

$p_{\mathrm{S}}^{\mathrm{AC},t}, p_{\mathrm{S}}^{\mathrm{DC},t}$	Amount of load shedding at AC and DC bus at time t [kW]
p_{PV}^{t}	Power output of PV at time t [kW]
p_{g}^{t}	Power output of generator g at time t [kW]
p_{UG}^{t}	Power of utility grid at time t [kW]
$p_{c}^{\mathrm{rec},t}$	Power of converter c from AC bus to DC bus at time t [kW]
$p_{c}^{\mathrm{inv},t}$	Power of converter c from DC bus to AC bus at time t [kW]
$p_{\mathrm{L},c}^{\mathrm{rec},t}, p_{\mathrm{L},c}^{\mathrm{inv},t}$	Conversion losses of converter c in the rectification and inversion mode at time t [kW]
$p_{\mathrm{ESS},v}^{\mathrm{ch},t}, p_{\mathrm{ESS},v}^{\mathrm{dc},t}$	Charging and discharging power of battery (connected to converter v) at time t [kW]
$p_{\mathrm{L},v}^{\mathrm{ch},t}, p_{\mathrm{L},v}^{\mathrm{dc},t}$	Conversion losses of converter v during the charging and discharging of batteries at time t [kW]
$e_{\mathrm{ESS},v}^{t}$	Energy stored in the battery at time t [kWh]
V^{DC}	DC bus voltage (V)

Neural Network

$\hat{\mathbf{z}}$	Sum of the weighted inputs and the bias of a neural network layer.
$\mathbf{z}$	Output vector of a neural network layer.
$\mathbf{W}, \mathbf{b}$	Weight matrix and bias vector of a neural network layer.
$\mathbf{z}_0, \mathbf{u}$	Input and output vectors of a neural network.
$\mathcal{J}$	The target function for neural network estimation.
$\mathcal{F}, h$	Equality and inequality constraints of the neural network input/output variables.
$\mathcal{L}$	Total neural network training loss.
$\mathcal{L}_{\mathcal{F}}, \mathcal{L}_h$	Neural network training loss caused by equality and inequality constraints.
$\mu_{\mathcal{F}}, \mu_h$	Penalty coefficients of the equality and inequality constraint losses.
$\beta_{\mathcal{F}}, \beta_h$	Growth factors of the penalty coefficients.
$\lambda_{\mathcal{F}}, \lambda_h$	Estimations of the Lagrangian multipliers of the equality and inequality constraint losses.
$\lambda_w, \lambda_{\mathrm{RS}}$	Regularization coefficient of the weight regularization and ReLU stability regularization terms.
L_k^j, U_k^j	Lower and upper bounds of $\hat{\mathbf{z}}$.
$I_{z,k}^j$	1 if the ReLU input is greater than or equal to zero, 0 otherwise.

I. Introduction

A MG IS a cluster of loads, storage systems, and microsources, operating as a single controllable system that capable of responding to central control signals [1], [2]. Hybrid AC/DC MGs provide efficient and flexible infrastructures to integrate various DC loads and renewable energy sources (RESs) into the existing grid, reducing dependence on fossil-fuel heavy power generations. Additionally, the integration of energy storage systems (ESSs) enables hybrid AC/DC MGs to achieve greater stability [3] and reliability [4]. In contrast to the individual AC or DC MGs, where DC-AC-AC-DC [5] or AC-DC-DC-AC [6], [7] power conversions are quite common, hybrid AC/DC MGs avoid the multiple power conversion stages by using bidirectional interlink converters (BICs).

Meanwhile, the integration of various loads and energy sources has brought new challenges to the energy management of hybrid AD/DC MGs. Due to the stochastic nature of RESs, the power balance between generation and consumption becomes less evident to achieve. In this regard, hybrid AC/DC MGs are often equipped with ESSs to have reliable backup power support, which involves additional DC/DC conversions. Moreover, the DC power demand is growing rapidly due to the excessive use of computers and network equipment in organizations such as universities and data centers [8]. As a result, these organizations require more AC/DC power conversions in their power supply. The cumulative effect of these power conversions renders the conversion losses significant in hybrid AC/DC MG energy management. To achieve more efficient operations of the energy management system (EMS), it is crucial to obtain precise approximations of the conversion losses. Traditional power [9], [10], [11], [12] or current [13], [14], [15] dependent conversion losses models are commonly used in EMSs but neglect the relationship between power losses and voltage. On the other hand, models that consider the voltage-dependent characteristic of the converters [16] are often too complex for integration into EMSs. Hence, a novel computational model should be proposed to quantify the impacts of variable voltage on the conversion losses and system efficiency while balancing precision and computational cost.

Analytical conversion loss surrogate modeling methods can be categorized into four classes:

1) Lossless modeling, which neglects conversion loss completely.
2) Proportional modeling, which assumes the conversion loss is proportional to the injected power.
3) Univariate modeling, which expresses the conversion loss as a quadratic function of converter current or power.
4) Multivariate modeling, which describes the conversion loss as a multivariate function of converter current/power and DC bus voltage.

A comparison of the analytical modeling methods in different studies is presented in Table I. Although all the mentioned modeling methods can be applied to both the AC/DC and DC/DC converters, most studies prefer proportional models for DC/DC converters. In contrast, the modeling of AC/DC converters varies more across studies.

In small isolated hybrid microgrids, the conversion losses are sometimes neglected entirely [17], [20], [23]. Proportional models are commonly employed for describing the DC/DC conversion loss of ESSs using fixed charging and discharging efficiencies [10], [15], [18], [19], [20], [21], [22]. For AC/DC conversion losses, proportional models often utilize separate AC-DC and DC-AC conversion efficiencies [10], [11], [18],

TABLE I
COMPARISON OF ANALYTICAL CONVERSION LOSS MODELS

Ref.	AC/DC Conversion Loss				DC/DC Conversion Loss			
	Lss	Prop	Uni	Mult	Lss	Prop	Uni	Mult
[16]				✓				
[9]			✓				✓	
[17]	✓				✓			
[18]		✓				✓		
[13]			✓					
[14]			✓		✓			
[19]			✓			✓		
[20]	✓					✓		
[21]		✓				✓		
[22]		✓				✓		
[23]	✓							
[11]		✓						
[15]			✓			✓		
[10]		✓				✓		
[24]			✓					

[1] Lss = lossless model; Prop = proportional model, or fixed conversion efficiency model; Uni = univariate current/power dependent model; Mult = multivariate model with both current and voltage dependent.

[21], [22]. Univariate models [9], [13], [14], [15], [18], [24] typically use three loss components with characteristic dependence on the converter current/power to approximate the conversion losses:

- Losses independent of the current/power, often caused by auxiliary systems.
- Losses linearly dependent on current/power, typically caused by switching losses.
- Losses quadratically dependent on current/power, typically caused by conduction losses.

Multivariate models account for voltage, current, and switching frequency to form a multivariable nonlinear conversion loss model for AC/DC conversion losses [16].

Numerous analytical conversion loss models have been proposed, primarily focusing on fixed-voltage DC buses. While MGs with fixed DC bus voltage simplify design and operation, the fixed voltage must be rated according to the maximum load power requirement [25]. This leads to the bus operating at a much higher voltage than necessary for a majority of operating conditions, incurring unnecessary power losses. Therefore, a variable-voltage DC bus with dynamic voltage control would allow the system to operate at a lower voltage, which would subsequently reduce the switching losses in converters. With the advent of variable-voltage DC buses in recent years [26], more efficient energy management schemes can leverage multivariate conversion loss models accounting for variable voltage.

Existing multivariate models used in EMSs are highly nonlinear. However, MILP is the most common optimization technique applied in EMSs [27], posing challenges in implementing efficient energy management with current conversion loss models. Artificial neural networks (ANNs or NNs) have demonstrated their excellent performance in function approximation [28], providing a potential solution for efficient surrogate modeling of highly nonlinear systems.

TABLE II
COMPARISON OF NN-BASED CONVERSION LOSS MODELS

Ref.	Model	AF/MF	# of Input Variables	Hard Constrained
[29]	ANFIS	Gaussian	3	✗
[30]	MLP	Not Mentioned	3	✗
[31]	MLP	Tanh	3	✗
[32]	MLP	Tanh	6	✗
[34]	PINN	Tanh	4	✗
[35]	CNN	ReLU	24×24	✗
[36]	LSTM	Tanh	> 100	✗
[37]	Transformer	Tanh	> 100	✗
This work	MLP	ReLU	2	✓

[1] AF = Activation Function; MF = Membership Function.

Prior studies have explored accurate NN-based models for conversion loss surrogate modeling. Techniques like adaptive neuro-fuzzy inference systems (ANFIS) have been applied to estimate the core losses of single-phase inverter transformers [29]. Sabanci et al. [30] showed that a simple multilayer perceptron (MLP) with only one hidden layer can estimate the DC/DC switching losses with small mean average errors (MAEs). Different NN architectures were tested in [31] for DC/DC loss prediction, different NN architectures were tested for DC/DC switching loss prediction, with an optimal design accurately estimating losses under varying switching frequencies. To reduce offline calibration needs, a self-adaptive NN for loss estimation was proposed in [32]. In recent years, physics-informed neural networks (PINNs) [33] have gained popularity in the field of surrogate modeling, with applications like DC-DC converter parameter estimation using fewer data to train than conventional data-driven models [34]. Other specialized NNs, such as CNN [35], LSTM [36] or Transformer [37], have been leveraged in core losses predictions incorporate different waveforms.

However, existing NN-based conversion loss models do not particularly suit for energy management problems. A comparative review of the existing NN-based conversion loss models is shown in Table II. It is observed that while the existing models can easily adopt multivariate formulation, most of them employ smooth activation functions/membership functions like hyperbolic tangent (tanh). The smooth activation functions in general lead to lower approximation errors, but also make the models not MILP compatible. While the CNN model with ReLU activation proposed in [35] can be reformulated into MILP constraints, the CNN architecture requires the input to be a 2D image form where the spacial features of adjacent pixels can be extracted by the convolutional layers. In energy management optimization problems often only a few factors related to the conversion loss are considered due to the computational cost, and thus the CNN-based method are not suitable for such problems. In addition, the existing NN-based conversion loss models do not take the hard constraints required by the MILP problems into account. The black-box nature of the NN-models may result in the models violating optimization constraints for certain operation conditions if

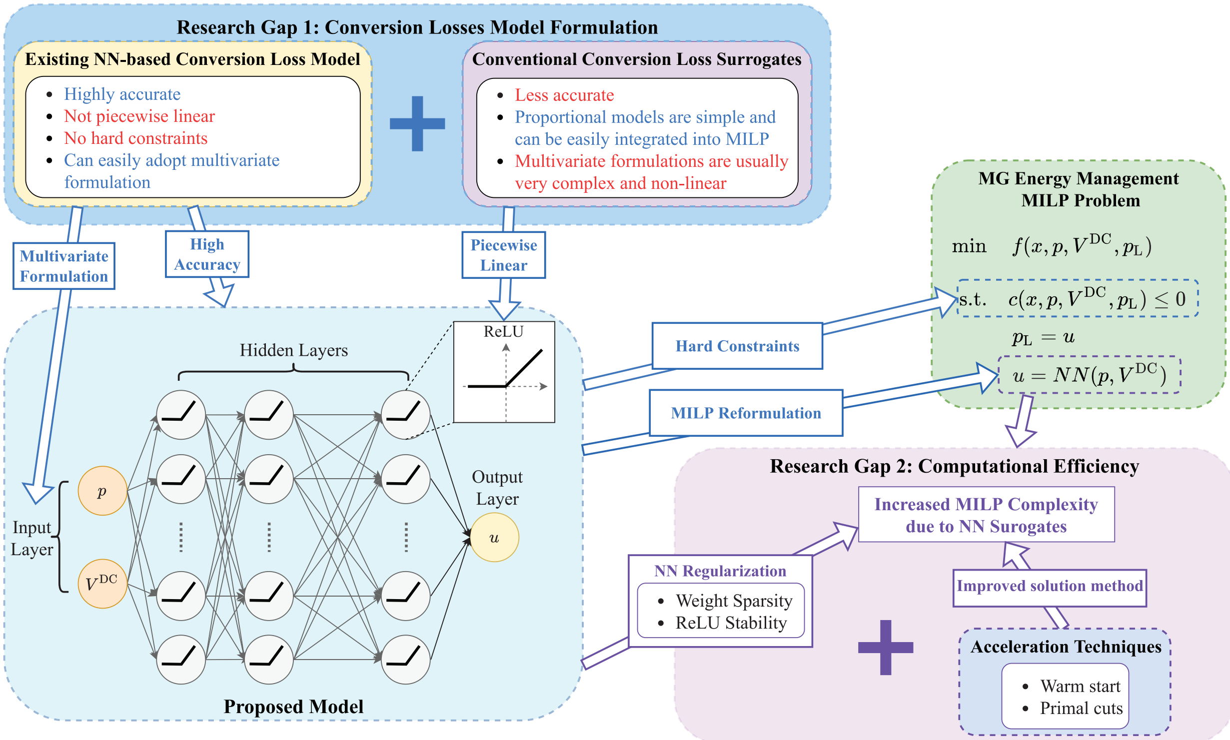

Fig. 1. Illustration of the research gaps and algorithmic novelty of the proposed method. Two research gaps are to be addressed in this work.

the constraints are not implicitly enforced in the training of NN-models. Furthermore, the Big-M reformulation of NN as MILP constraints increases the computation cost of the MILP problems significantly. As a result, the application of NN-based techniques in EMSs has primarily focused on forecasting [38], [39] and optimization [40], [41], [42], [43] objectives. There has been a lack of study in piecewise linear NN surrogates tailored for MILP problems.

To address the identified research gaps, we propose an NN-based multivariate conversion loss surrogate model for more efficient energy management, which is formulated as a two-stage stochastic MILP problem. The NN model utilizes ReLU activation, enabling exact MILP reformulation. Furthermore, optimization constraints are incorporated into the NN model during training to improve feasibility. To manage the computational cost associated with NN surrogates, we have developed a workflow that optimizes both the NN weights and the MILP solution method. Fig. 1 gives a visual illustration of the research gaps and the novelties of the proposed approach.

In summary, the key contributions of this work are:

- Unlike existing conversion loss surrogate models that assume fixed DC bus voltage, our model considers conversion losses as multivariate functions with respect to converter power and voltage. We develop a NN-based linear surrogate model that approximates these functions as MILP constraints for energy management problems. By incorporating the DC bus voltage as decision variables, the total conversion loss can be further optimized in energy management problems.
- To tailor the NN model for MILP problems, the NN model is trained with additional hard constraints. We enforce physical and statistical constraints on the NN input/output variables during training using the augmented Lagrangian method. Notably, a relative error constraint is introduced during the NN training process, acting as both an attention mechanism for the NN model and a solution acceleration technique for the optimization problem.
- To minimize computational costs associated with the proposed surrogate model, we optimize NN model weights through weight pruning and ReLU stability regularization. The MILP solution method is also enhanced by adopting a warm start strategy and employing linear and primal cuts. These improvements address the increased computational burden resulting from NN surrogates.
- To validate the proposed surrogate model and solution method, we formulate a two-stage stochastic UC problem with a variable-voltage DC bus. The proposed surrogate model is integrated into the UC problem as linear constraints with mixed-integer variables. Results demonstrate that the UC problem can be efficiently solved due to NN weight regularization and the improved solution method. Comparative case studies against fixed and current-dependent conversion efficiency models indicate the effectiveness of using a variable-voltage DC bus in reducing total energy loss in hybrid MGs. Finally, hardware-in-the-loop experiments were conducted to further validate the proposed model.

The rest of this paper is structured as follows: We begin in Section II by giving an overview of efficient energy management of hybrid AC/DC MGs. The proposed neural network-based linear surrogate model is presented in

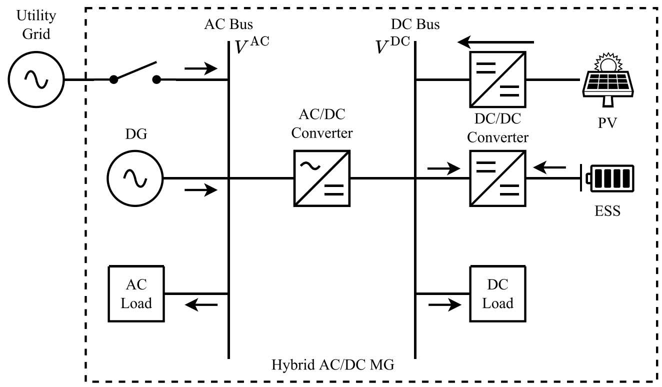

Fig. 2. Structure overview of a typical hybrid AC/DC MG.

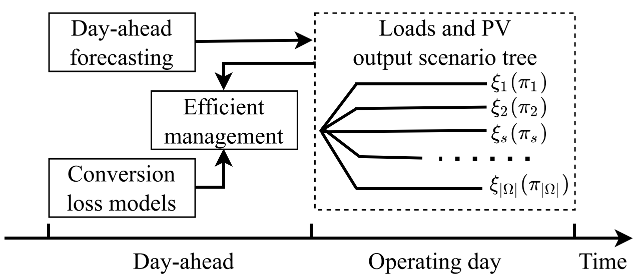

Fig. 3. Efficient management of networked hybrid AC/DC MGs. $\xi_s :=$ {load profile, PV output profile}. π_s is the corresponding probability of scenario s. The scenario tree is derived from the day-ahead forecasting and treated as an input for the proposed efficient management.

Section III. Section IV introduces the two-stage stochastic UC problem and its solution method. Section V illustrates the results of the case studies. Conclusions and future work are highlighted in Section VI.

II. Efficient Management of Hybrid AC/DC MGs

As shown in Fig. 2, consider a typical hybrid AC/DC MG composed of one AC bus and one DC bus, the terminal voltage magnitudes of which are depicted by V_{AC} and V_{DC}, respectively. The MG is interconnected to the utility grid through the AC bus. The diesel generators (DGs) and AC loads are also integrated into the AC bus. DC sources and loads are connected to the DC bus, including ESSs, renewable energy sources like photovoltaic (PV) generators, and DC loads. Each DC source is equipped with a bi-directional DC/DC converter $v \in \mathcal{V}$. The AC and DC buses are connected through a bi-directional AC/DC converter $c \in \mathcal{C}$.

In this work, efficient management is implemented in the day-ahead operation, as shown in Fig. 3. Before the operating day, the efficient management is implemented after the forecasting of loads and PV output profile along the operating day. The forecasting results are depicted as a scenario tree $\xi_s \in \Omega$ with a given probability density function $\pi_s, \forall s \in \Omega$.

III. Neural Network-Based Linear Surrogate Model of Conversion Losses

This section discusses the details of the neural network-based surrogate modeling workflow, including the data generation, the neural network training process incorporating hard constraints, and the MILP reformulation of the neural network model.

A. Data Generation Using Analytical Models

Surrogate models are data-driven models that approximate the output of more complex models based on given parameters [44]. In this study, the goal of the proposed surrogate modeling method is to accurately approximate the conversion losses of the bi-directional AC/DC and DC/DC converters. The converter losses can be written as a multivariate function w.r.t. the converter power and DC bus voltage:

$$p_L = \mathcal{J}\left(p, V^{\mathrm{DC}}\right) \tag{1}$$

The multivariable, nonlinear analytical models of both the AC/DC and DC/DC converters proposed by [45] are employed to generate high-fidelity sample data for the surrogate model optimization. Four datasets of size 100000 were generated by the analytical models for rectification, inversion, charging and discharging losses of the converters, respectively. Each dataset was then randomly split into training, validation and test datasets with a ratio of 7:2:1. The data generation details and the generated datasets are publicly available online at [46].

The simulated conversion loss coefficient curves are shown in Fig. 4. The conversion loss coefficient η_{L} is defined as:

$$\eta_{\mathrm{L}} = 1 - \eta = \frac{p_L}{p} \times 100\% \tag{2}$$

where p is the input power and η is the conversion efficiency:

$$\eta = \frac{p - p_{\mathrm{L}}}{p} \times 100\% \tag{3}$$

The conversion loss coefficient curves indicate that the conversion losses can be reduced by lowering the DC bus voltage. Therefore a multivariate surrogate model utilizing the variable-voltage DC bus could result in lower total conversion losses in EMSs.

B. Non-Linear Conversion Losses Approximations Using Neural Networks

NNs have been widely used in surrogate modeling due to their excellent approximation capabilities. Feed-forward NNs are proven to be capable of approximating any smooth function on a compact domain with arbitrary precision given a sufficient number of neurons [47]. As depicted in Fig. 5, a fully connected neural network (FCNN) can be represented as a directed acyclic graph with nodes structured into connected layers. The first and the last layers are the input and output layers, respectively. In between the input and output layers are the hidden layers. Let $\mathbf{z}_0$ be the input vector of the neural network, $\mathbf{z}_k$ be the output vector of the kth hidden layer. The output $\mathbf{u}$ of a neural network with K hidden layers is given by:

$$\hat{\mathbf{z}}_{k+1} = \mathbf{W}_{k+1}\mathbf{z}_k + \mathbf{b}_{k+1} \quad \forall k = 0, 1, \ldots, K-1, \tag{4}$$

$$\mathbf{z}_k = \sigma\left(\hat{\mathbf{z}}_k\right), \quad \forall k = 1, 2, \ldots, K, \tag{5}$$

$$\mathbf{u} = \mathbf{W}_{K+1}\mathbf{z}_K + \mathbf{b}_{K+1}, \tag{6}$$

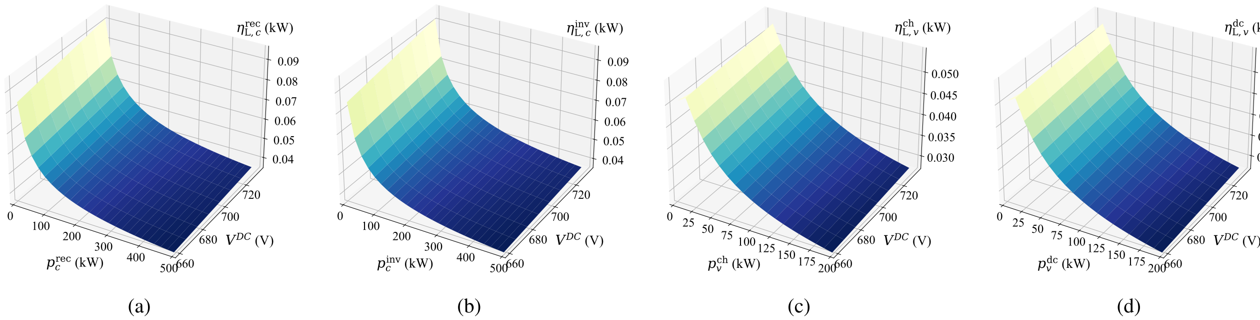

Fig. 4. Conversion loss coefficient of different modes of the AC/DC and DC/DC converters under different voltage magnitude and power levels. (a) AC/DC Rectification; (b) AC/DC Inversion; (c) DC/DC Charging; (d) DC/DC Discharging.

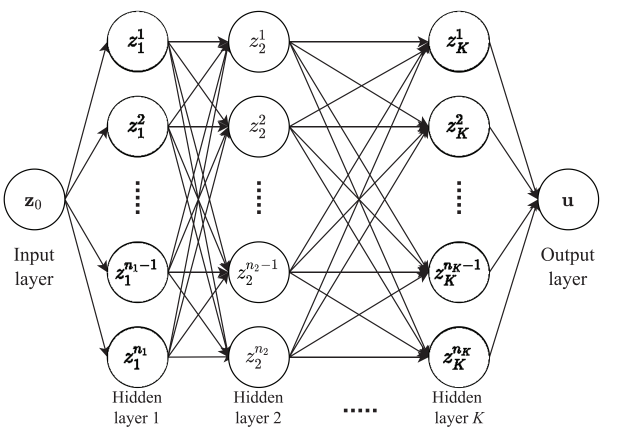

Fig. 5. Illustration of a fully connected neural network structure with K hidden layers.

where σ is a nonlinear activation function, and $\mathbf{W}_k$, $\mathbf{b}_k$ are the weight matrix and bias vector of the kth layer, respectively.

The nonlinear activation makes it possible for FCNNs to approximate nonlinear functions. However, many common activation functions, such as hyperbolic tangent or sigmoid, can not be expressed precisely in MILP problems. In order to embed neural networks in MILP problems, it is necessary to use a piecewise activation function like the ReLU function. In this work, we only discuss FCNNs with ReLU activation function, whereas the proposed method can be easily applied to FCNNs with other piecewise linear activation functions. The ReLU activation function is formulated as:

$$\mathrm{ReLU}(\hat{\mathbf{z}}) = \max(\hat{\mathbf{z}}, 0). \tag{7}$$

In this work, FCNNs is employed to approximate the conversion losses:

$$\mathbf{u}(\mathbf{z}_0) \approx \mathcal{J}(\mathbf{z}_0) = p_{\mathrm{L}} \tag{8}$$

where $\mathbf{z}_0 = [p, V^{\mathrm{DC}}]^T$ is the input vector of the neural network.

The weight matrices $\mathbf{W}$ and bias vectors $\mathbf{b}$ of the neural network are adjustable parameters that need to be optimized based on the training data. Let θ be the set of all trainable parameters of an FCNN. For function approximation tasks, the FCNN is usually trained by minimizing the mean square error (MSE) between the network output and the training samples:

$$\min_{\theta}\ \mathcal{L}_{\mathcal{J}} = \frac{1}{N}\sum_{i=1}^{N}\left|\mathbf{u}\left(\mathbf{z}^i_{0,\mathrm{train}}\right) - \mathbf{u}^i_{\mathrm{train}}\right|^2, \tag{9}$$

where $\{\mathbf{z}_{0,\mathrm{train}}=[p, V_{\mathrm{DC}}]^T, \mathbf{u}_{\mathrm{train}}=[p_{\mathrm{L}}]\}_{i=1}^N$ is the training dataset consists of input/output pairs of the multivariate conversion loss function.

In energy management problems, the input and output of the surrogate models often correspond to some of the decision variables, which are limited by the problem constraints. Unlike the MSE loss, which is a soft constraint, the constraints on the decision variables are hard constraints that need to be satisfied to make the solution feasible. While the basic MSE-based training method could produce models with small errors, the input and output of the neural networks are not guaranteed to satisfy the hard constrained required by the energy management problems. Thus, pure data-driven surrogate modeling approaches could potentially produce output variables that violate the optimization constraints when the input points are unseen in the training dataset.

C. Training Neural Networks With Hard Constraints

When additional hard constraints are added to the neural network, the learning problem then becomes:

$$\begin{aligned}
\min_{\theta}\ & \mathcal{L}_{\mathcal{J}} = \frac{1}{N}\sum_{i=1}^{N}\left|\mathbf{u}\left(\mathbf{z}^i_{0,\mathrm{train}}\right) - \mathbf{u}^i_{\mathrm{train}}\right|^2, \\
\text{s.t. } & \mathcal{F}[\mathbf{z}_0, \mathbf{u}] = 0, \\
& h[\mathbf{z}_0, \mathbf{u}] \le 0,
\end{aligned} \tag{10}$$

where $\mathcal{F}$ and h are the additional equality and inequality constraints, respectively.

In the proposed energy management problem, we consider the following additional constraints on the surrogate input and output variables:

$$0 \le \mathbf{u}(\mathbf{z}_0) \le p, \tag{11}$$

$$\frac{1}{N}\sum_{i=1}^{N}\left|\frac{\mathbf{u}\left(\mathbf{z}^i_{0,\mathrm{train}}\right) - \mathbf{u}^i_{\mathrm{train}}}{\mathbf{u}^i_{\mathrm{train}}}\right| \le \epsilon, \tag{12}$$

where ϵ is a threshold on the training relative error.

Constraint (11) limits the range of the output conversion losses. Since the power loss curve is highly nonlinear when the converter power p is small, the relative error of the surrogate model will be large if the model fails to capture the nonlinearity very well. Therefore (12) operates as an attention mechanism, facilitating the model in capturing the intricacies of the highly nonlinear region within the power loss curve. In the case studies, it is observed that a large portion of the solutions lies in the area where the power loss are 0 or near 0. Constraint (12) plays a pivotal role not only in enhancing the model's understanding of the nonlinearities but also in assisting the MILP solver in determining optimal solutions, particularly in cases where power losses are close to or equal to zero.

Commonly used NN training methods, such as stochastic gradient descent (SGD), are not able to solve constrained optimization problems. One possible approach to enforce the additional constraints using the existing training methods is to convert the constrained optimization problems to unconstrained problems by adding additional penalty terms to the training loss function:

$$\min_{\theta} \ \mathcal{L} = \mathcal{L}_{\mathcal{J}} + \mu_{\mathcal{F}}\mathcal{L}_{\mathcal{F}} + \mu_h \mathcal{L}_h, \tag{13}$$

where $\mu_{\mathcal{F}} > 0$ and $\mu_h > 0$ are the fixed penalty coefficients, and $\mathcal{L}_{\mathcal{F}} = \mathcal{F}^2[\mathbf{z}_0, \mathbf{u}]$, $\mathcal{L}_h = (\max(h[\mathbf{z}_0, \mathbf{u}], 0))^2$ are the quadratic penalties for the equality and inequality constraints, respectively. However, this approach may lead to ill-conditioning if the penalty coefficients are too large. On the other hand, the model will not satisfy the hard constraints very well if the penalty coefficients are not large enough. We can utilize the augmented Lagrangian method to avoid the ill-condition problem in the training process. The augmented Lagrangian method decomposes the original constrained optimization problem into a series of unconstrained optimization problems using the Lagrangian multipliers. The unconstrained optimization problem at iteration k is given as:

$$\min_{\theta} \ \mathcal{L}^k = \mathcal{L}_{\mathcal{J}} + \mu^k_{\mathcal{F}}\mathcal{L}_{\mathcal{F}} + \lambda^k_{\mathcal{F}}\mathcal{F}[\mathbf{z}_0, \mathbf{u}] + \mu^k_h \mathcal{L}_h + \lambda^k_h h[\mathbf{z}_0, \mathbf{u}], \tag{14}$$

where $\lambda_{\mathcal{F}}$ and λ_h are estimations of the Lagrange multipliers. Following [48], [49], in each iteration the penalty coefficients and the multiplier variables are updated by:

$$\mu^{k+1}_{\mathcal{F}} = \beta_{\mathcal{F}}\mu^k_{\mathcal{F}}, \tag{15}$$
$$\mu^{k+1}_h = \beta_h \mu^k_h, \tag{16}$$
$$\lambda^{k+1}_{\mathcal{F}} = \lambda^k_{\mathcal{F}} + 2\mu^k_{\mathcal{F}}\mathcal{F}[\mathbf{z}_0, \mathbf{u}], \tag{17}$$
$$\lambda^{k+1}_h = \lambda^k_h + 2\mu^k_h h[\mathbf{z}_0, \mathbf{u}]. \tag{18}$$

The pseudocode of the training process using the augmented Lagrangian method is presented in Algorithm 1.

D. Exact MILP Reformulation for ReLU Neural Networks

A properly trained ReLU neural network can be expressed as a combination of affine transformation and piecewise linear activations. Utilizing the exact reformulation of ReLU activation [50], an MILP reformulation of the network is given by:

Algorithm 1: Hard Constrained Neural Network Training via the Augmented Lagrangian Method

Data: initial penalty coefficients $\mu^0_{\mathcal{F}}$ and μ^0_h, factors $\beta_{\mathcal{F}}$ and β_h, training dataset $[\mathbf{z}_{0,\text{train}}, \mathbf{u}_{\text{train}}]_{i=1}^N$
Result: Trained neural network $\mathbf{u}(\theta_u^*)$
Set $k = 0$, $\lambda^0_{\mathcal{F}} = 0$ and $\lambda^0_h = 0$, train the neural network from random initialization until the training loss is converged: $\theta^0_u = \min_{\theta} \mathcal{L}^0(\theta_u)$
while *$\mathcal{L}_{\mathcal{F}}$ and $\mathcal{L}_h$ are greater than a tolerance* **do**
    $\mu^{k+1}_{\mathcal{F}} = \beta_{\mathcal{F}}\mu^k_{\mathcal{F}}$
    $\mu^{k+1}_h = \beta_h \mu^k_h$
    $\lambda^{k+1}_{\mathcal{F}} = \lambda^k_{\mathcal{F}} + 2\mu^k_{\mathcal{F}}\mathcal{F}[\mathbf{z}_0, \mathbf{u}]$
    $\lambda^{k+1}_h = \lambda^k_h + 2\mu^k_h h[\mathbf{z}_0, \mathbf{u}]$
    Train the neural network from θ^k_u until the training loss is converged: $\theta^{k+1}_u = \min_{\theta} \mathcal{L}^{k+1}(\theta_u)$ $k = k + 1$
end

(4), (6),

$$z^j_k \le \hat{z}^j_k - L^j_k\left(1 - I^j_{z,k}\right), \tag{19}$$
$$z^j_k \ge \hat{z}^j_k, \tag{20}$$
$$z^j_k \le U^j_k I^j_{z,k}, \tag{21}$$
$$z^j_k \ge 0, \tag{22}$$
$$I^j_{z,k} \in \{0, 1\}, \tag{23}$$
$$L^j_k \le \hat{z}^j_k \le U^j_k, \tag{24}$$

where L^j_k and U^j_k are required to bound $\hat{\mathbf{z}}^j$. This reformulation introduces an additional binary variable $I^j_{z,k}$ to each neuron. In standard MILP solvers, it is desirable for the users to provide tight variable bounds to reduce the solution time. Interval arithmetic yields valid bounds for $k \ge 1$:

$$U^j_k = \sum_{i=1}^{n_{k-1}} \max\left\{w^{ji}_k \max\left\{0, U^i_{k-1}\right\}, w^{ji}_k \max\left\{0, L^i_{k-1}\right\}\right\} + b^j_k, \tag{25}$$

$$L^j_k = \sum_{i=1}^{n_{k-1}} \min\left\{w^{ji}_k \max\left\{0, U^i_{k-1}\right\}, w^{ji}_k \max\left\{0, L^i_{k-1}\right\}\right\} + b^j_k, \tag{26}$$

The input vector $\mathbf{z}_0$ is often normalized before being fed into the neural network to improve the back propagation dynamics:

$$\mathbf{z}_0 = \left[\frac{p - L_p}{U_p - L_p}, \frac{V^{\text{DC}} - L_{V^{\text{DC}}}}{U_{V^{\text{DC}}} - L_{V^{\text{DC}}}}\right]^T \tag{27}$$

where U_p, $U_{V^{\text{DC}}}$ are the upper boundary of p and V^{DC}, and L_p, $U_{V^{\text{DC}}}$ are the lower boundary of p and V^{DC}. Since every input dimension is normalized to be in range [0, 1], we have $U^j_0 = 0$ and $L^j_0 = 1$ for every input dimension.

E. Improving MILP Solving Speed With Weight Sparsity and ReLU Stability

Although the proposed hard-constrained surrogate model can provide accurate and feasible approximations, the computational complexity of the associated MILP constraints is not optimized. MILP systems that adopt the proposed surrogate model would benefit from fewer decision variables. The number of decision variables in the MILP reformulation can be reduced by optimizing the weight sparsity and stability of the proposed NN model during training. In this study, we applied the following methods to enforce weight sparsity and stability:

1) L1 Weight Regularization: Weight regularization is a common technique used in NN training to prevent overfitting. L1 and L2 regularizations are the most popular regularization methods. Both of them penalize large weight values, which are usually signs of overfitting. However, L1 regularization tends to result in models with more sparse weights [51]. Therefore, we adopt L1 regularization during training to improve weight sparsity.

2) Weight Pruning: We applied two types of weight pruning to the proposed model:

1) Fixed sparsity pruning: A automated gradual pruning algorithm proposed in [52] is also adopted to enforce a certain level of sparsity. Starting from 0% sparsity, a defined percentage of weights is set to 0 based on absolute magnitude. The NN model is re-trained with the updated structure. By repeating this process, the sparsity target can be achieved gradually.
2) Small weights pruning: Zeroing out weights based on a threshold value. In this study, we verified on the test dataset that zeroing out weights that are less than 1e-4 has no noticeable impact on model performance.

3) ReLU Stability: The reformulation described in Section III-D adds a binary variable to each neuron to determine the activation status of the corresponding ReLU function. However, based on the bounds of $\hat{\mathbf{z}}$ obtained from the bound tightening algorithm, we can predetermine if a ReLU function will be activated. If $U \leq 0$, the ReLU function will never be activated; likewise, the ReLU function will always be activated if $L \geq 0$. We call a ReLU is *stable* if it is always activated or inactivated. A stable ReLU can be replaced with a zero function or an identity function, and the corresponding variables $\hat{\mathbf{z}}$ and I can be removed from the reformulation. Therefore, increasing the number of stable ReLUs in the model can effectively reduce the number of decision variables in the reformulated MILP constraints. The stability of a ReLU function can be verified based on the signs of the bounds: If the signs of U and L are the same, the ReLU is stable; otherwise, the ReLU may be unstable. Ideally, we want to use the following indicator function as a regularizer in the loss function to improve the ReLU stability of the model:

$$F_{\mathrm{RS}}(U, L) = \mathrm{sign}(U) \cdot \mathrm{sign}(L) \tag{28}$$

However, it's difficult to directly use the signs of the bounds as a regularizer for the NN model since the commonly used sign function is non-differentiable. Instead, we can use a smooth approximation of F_{RS} [53]:

$$F^*_{\mathrm{RS}}(U, L) = -\tanh(1 + U \cdot L) \tag{29}$$

The final loss function after applying L1 regularization and ReLU stability is given by:

$$\begin{aligned}\min_{\theta}\ \mathcal{L}^k = {} & \mathcal{L}_{\mathcal{J}} + \mu^k_{\mathcal{F}}\mathcal{L}_{\mathcal{F}} + \lambda^k_{\mathcal{F}}\mathcal{F}[\mathbf{z}_0, \mathbf{u}] \\ & + \mu^k_h\mathcal{L}_h + \lambda^k_h h[\mathbf{z}_0, \mathbf{u}] \\ & + \lambda_w \sum_{w\in\mathbf{W}} |w| + \lambda_{\mathrm{RS}} F^*_{\mathrm{RS}}(U, L),\end{aligned} \tag{30}$$

where λ_w is the L1 regularization coefficient for the weight values and λ_{RS} is the coefficient of the ReLU stability regularizer.

IV. Two-Stage Stochastic Unit Commitment

A. Two-Stage Stochastic Unit Commitment Under Linear Surrogate Losses

The UC problem for a hybrid AC/DC MG has been formulated as the following two-stage stochastic programming problem:

$$\min_{\mathbf{x}\in\mathcal{X}} f(\mathbf{x}) + \sum_{\xi_s\in\Omega} \pi_s[\mathcal{Q}(\mathbf{x}, \xi_s)], \tag{31}$$

$$\mathcal{Q}(\mathbf{x}, \xi_s) = \min_{\mathbf{y}_s\in\mathcal{Y}(\mathbf{x},\xi_s)} g(\mathbf{y}_s), \tag{32}$$

where $\mathbf{x} := \{\alpha^t_g, \beta^t_g, u^t_g, \forall g, t\}$ is a binary vector, $\mathcal{X}$ is the first-stage constraint set. Ω is the uncertainty set of random vector $\xi_s := \{p^{\mathrm{AC},t}_{\mathrm{D}}, p^{\mathrm{DC},t}_{\mathrm{D}}, \overline{P}^t_{\mathrm{PV}}, \forall t\}$, including the loads and PV output. $\mathcal{Q}(\mathbf{x}, \xi_s)$ is the dynamic OPF with given $\mathbf{x}$ and ξ_s, as an mixed-integer linear programming (MILP) problem, where $\mathbf{y}_s := \{p^t_g, p^t_{\mathrm{UG}}, p^{\mathrm{rec},t}_c, p^{\mathrm{inv},t}_c, p^{\mathrm{rec},t}_{\mathrm{L},c}, p^{\mathrm{inv},t}_{\mathrm{L},c}, p^{\mathrm{ch},t}_{\mathrm{ESS},v}, p^{\mathrm{dc},t}_{\mathrm{ESS},v}, p^{\mathrm{ch},t}_{\mathrm{L},v}, p^{\mathrm{dc},t}_{\mathrm{L},v}, e^t_{\mathrm{ESS},v}, I^t_{\mathrm{ESS},v}, I^t_c, \forall g, t\}$ is a mixed-integer vector, due to existence of the binary variables $I^t_{\mathrm{ESS},v}$ and I^t_c. $\mathcal{Y}(\mathbf{x}, \xi_s)$ is a convex set, where the power losses $p^{\mathrm{rec},t}_{\mathrm{L},c}, p^{\mathrm{inv},t}_{\mathrm{L},c}, p^{\mathrm{ch},t}_{\mathrm{L},v}$ and $p^{\mathrm{dc},t}_{\mathrm{L},v}$ are depicted by their corresponding linear surrogate models.

B. First-Stage Optimization

The first-stage is to minimize the start-up and shut-down cost of DGs, by optimizing their start-up, shut-down and operating status, as follows:

$$\min_{\mathbf{x}\in\mathcal{X}} f(\mathbf{x}) = \sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}} \left(c_{\mathrm{U},g}\alpha^t_g + c_{\mathrm{D},g}\beta^t_g\right) \tag{33}$$

The first-stage constraint set $\mathcal{X}$ includes:

$$\alpha^t_g - \beta^t_g = u^t_g - u^{t-\Delta t}_g, \forall t, g \tag{34}$$

$$\sum_{q=t-\underline{\mathrm{UT}}_g+1}^{t} \alpha^q_g \leq u^t_g, \forall t \in \{\underline{\mathrm{UT}}_g, \ldots, T\}, g \tag{35}$$

$$\sum_{q=t-\underline{\mathrm{DT}}_g+1}^{t} \beta^q_g \leq 1 - u^t_g, \forall t \in \{\underline{\mathrm{DT}}_g, \ldots, T\}, g \tag{36}$$

$$\sum_{q=t}^{t+\overline{\mathrm{UT}}_g} u^q_g \leq \overline{\mathrm{UT}}_g, \forall t \in \{1, \ldots, T - \overline{\mathrm{UT}}_g\}, g \tag{37}$$

The operating status transition constraint is given by Eqn. (34). Eqns. (35) and (36) depict the minimum start-up and shut-down duration constraints, respectively. The maximum up duration is constrained by Eqn. (37).

C. Second-Stage Optimization

The second-stage optimization is to minimize the total power losses, via optimizing the power output of DGs and ESSs, the voltage of AC/DC converters and DC/DC converters and etc., as the following dynamic OPF problem:

$$\min_{\mathbf{y}_s\in\mathcal{Y}(\mathbf{x},\xi_s)} g(\mathbf{y}) = \sum_{t\in\mathcal{T}}\Bigg[c_{\mathrm{L}}\sum_{c\in\mathcal{C}}\Big(p_{\mathrm{L},c}^{\mathrm{rec},t}+p_{\mathrm{L},c}^{\mathrm{inv},t}\Big)+c_{\mathrm{L}}\sum_{v\in\mathcal{V}}\Big(p_{\mathrm{L},v}^{\mathrm{ch},t}+p_{\mathrm{L},v}^{\mathrm{dc},t}\Big) + c_{\mathrm{S}}\Big(p_{\mathrm{S}}^{\mathrm{AC},t}+p_{\mathrm{S}}^{\mathrm{DC},t}\Big)\Bigg] \tag{38}$$

The second-stage constraint set $\mathcal{Y}(\mathbf{x},\xi)$ includes both MGs and distribution systems constraints.

$$p_{\mathrm{UG}}^t+\sum_{g\in\mathcal{G}}p_g^t+\sum_{c\in\mathcal{C}}\Big(p_c^{\mathrm{inv},t}-p_{\mathrm{L},c}^{\mathrm{inv},t}\Big) = p_{\mathrm{D}}^{\mathrm{AC},t}-p_{\mathrm{S}}^{\mathrm{AC},t}+\sum_{c\in\mathcal{C}}p_c^{\mathrm{rec},t}, \forall t \tag{39}$$

$$\sum_{v\in\mathcal{V}}\Big(p_{\mathrm{ESS},v}^{\mathrm{dc},t}-p_{\mathrm{ESS},v}^{\mathrm{ch},t}\Big)+\sum_{c\in\mathcal{C}}\Big(p_c^{\mathrm{rec},t}-p_{\mathrm{L},c}^{\mathrm{rec},t}\Big)+p_{\mathrm{PV}}^t = p_{\mathrm{D}}^{\mathrm{DC},t}-p_{\mathrm{S}}^{\mathrm{DC},t}+\sum_{c\in\mathcal{C}}p_c^{\mathrm{inv},t}, \forall t \tag{40}$$

$$\underline{P}_g u_g^t \le p_g^t \le \overline{P}_g u_g^t, \forall t, g \tag{41}$$

$$p_g^{t-\Delta t}-p_g^t \le R_g^+\Delta t, \forall t, g \tag{42}$$

$$p_g^t-p_g^{t-\Delta t} \le R_g^-\Delta t, \forall t, g \tag{43}$$

$$0 \le p_{\mathrm{UG}}^t \le \overline{P}_{\mathrm{UG}}, \forall t \tag{44}$$

$$0 \le p_{\mathrm{PV}}^t \le \overline{P}_{\mathrm{PV}}^t, \forall t \tag{45}$$

$$0 \le p_{\mathrm{S}}^{\mathrm{AC},t} \le p_{\mathrm{D}}^{\mathrm{AC},t}, \forall t \tag{46}$$

$$0 \le p_{\mathrm{S}}^{\mathrm{DC},t} \le p_{\mathrm{D}}^{\mathrm{DC},t}, \forall t \tag{47}$$

$$0 \le p_c^{\mathrm{rec},t} \le \overline{P}_c I_c^t, \forall t, c \tag{48}$$

$$0 \le p_c^{\mathrm{inv},t} \le \overline{P}_c\big(1-I_c^t\big), \forall t, c \tag{49}$$

$$0 \le p_{\mathrm{ESS},v}^{\mathrm{ch},t} \le \overline{P}_{\mathrm{ESS},v}^{\mathrm{ch}} I_{\mathrm{ESS},v}^t, \forall t, v \tag{50}$$

$$0 \le p_{\mathrm{ESS},v}^{\mathrm{dc},t} \le \overline{P}_{\mathrm{ESS},v}^{\mathrm{dc}}\big(1-I_{\mathrm{ESS},v}^t\big), \forall t, v \tag{51}$$

$$e_{\mathrm{ESS},v}^t = e_{\mathrm{ESS},v}^{t-\Delta t}+\Big(p_{\mathrm{ESS},v}^{\mathrm{ch},t}-p_{\mathrm{L},v}^{\mathrm{ch},t}-p_{\mathrm{ESS},v}^{\mathrm{dc},t}-p_{\mathrm{L},v}^{\mathrm{dc},t}\Big)\Delta t, \forall t \tag{52}$$

$$\underline{E}_{\mathrm{ESS},v} \le e_{\mathrm{ESS},v}^t \le \overline{E}_{\mathrm{ESS},v}, \forall t, v \tag{53}$$

The AC and DC bus power balance constraints are shown in Eqns.(39) and (40), respectively. The generator ramp and output constraints are given by Eqns. (41)-(43). The power exchange between the MG and utility grid is limited by Eqn. (44). Eqn. (45) is the constraint for PV output. The load shedding on AC and DC bus are constrained by Eqns. (46) and (47), respectively. The power conversion constraints on the interlinking converters are illustrated in Eqns. (48) and (49). The energy and power related constraints of battery ESSs are depicted by Eqns. (50)-(53). In Eqns. (39), (40) and (52), the power losses, i.e., $p_{\mathrm{L},c}^{\mathrm{rec},t}$, $p_{\mathrm{L},c}^{\mathrm{inv},t}$, $p_{\mathrm{L},v}^{\mathrm{ch},t}$, and $p_{\mathrm{L},v}^{\mathrm{dc},t}$, are non-convex w.r.t. voltage and power. Using the proposed linear surrogate models (4), (6), (19) - (24), these losses are depict by a set of linear constraints with mixed-integer variables.

D. Solution Method

1) Convergence Guarantee With Improved Linear Cuts: Bender's decomposition algorithm [54] is a powerful technique for efficiently solving large-scale two-stage stochastic problems. By decomposing the MILP problem (31) into a master problem (33) and a set of subproblems (38), it enables parallel processing of subproblems, significantly reducing computational costs. However, due to the presence of binary variables in $\mathbf{y}s$, the traditional duality-based Benders decomposition may not guarantee algorithm convergence and solution optimality. To address this, improved linear cuts [55] are utilized to approximate the mixed recourse and ensure algorithm convergence. The linear cuts are obtained through two convex optimization problems. First, we solve (54) for the Lagrange multiplier λ_s of the equality constraint $\hat{\mathbf{x}} = \mathbf{x}^*$ for a given first-stage solution $\mathbf{x}^*$:

$$\begin{aligned}\hat{\mathcal{Q}}\big(\hat{\mathbf{x}},\xi_s\big) = \min_{\hat{\mathbf{x}},\mathbf{y}_s}\ & g(\mathbf{y}_s)\\ \text{s.t. Eqns. } & (39)-(53),\\ & \hat{\mathbf{x}} = \mathbf{x}^* : (\lambda_s)\end{aligned} \tag{54}$$

With the obtained λ_s, the following problem is solved to estimate the lower boundary of $\hat{\mathcal{Q}}(\mathbf{x}^*,\xi_s)$ under ξ_s:

$$\begin{aligned}\mathcal{Q}'\big(\tilde{\mathbf{x}},\xi_s\big) = \min_{\tilde{\mathbf{x}},\mathbf{y}_s}&\big[g(\mathbf{y}_s)-\lambda_s^{\mathrm{T}}\tilde{\mathbf{x}}\big]\\ \text{s.t. Eqns. } & (39)-(53)\\ & \tilde{\mathbf{x}}\in\mathbb{R}^{n1}, \quad \mathbf{y}_s\in\mathbb{B}^{n_2}\times\mathbb{R}^m\end{aligned} \tag{55}$$

The derived λ_s and $\hat{\mathcal{Q}}(\mathbf{x}^*,\xi_s)$ are then used to formulate the linear cuts under ξ_s:

$$\theta_s \ge \lambda_s^{\mathrm{T}}\mathbf{x}+\mathcal{Q}'\big(\tilde{\mathbf{x}},\xi_s\big), \tag{56}$$

which are integrated into the master problem as constraints:

$$\begin{aligned}&\min_{\mathbf{x},\theta_s}\ f(\mathbf{x})+\sum_{s=1}^N\pi_s\theta_s\\ &\text{s.t. } (34)-(37),\\ &\quad \theta_s \ge \lambda_s^{l,\mathrm{T}}\mathbf{x}+\mathcal{Q}'^{l}\big(\tilde{\mathbf{x}},\xi_s\big), \quad l=1,\ldots,k\\ &\quad \mathbf{x}\in\mathbb{B}^{n_1}, \theta_s\in\mathbb{R}, \forall s\end{aligned} \tag{57}$$

In each iteration, the lower bound (LB) of the two-stage problem is estimated by $f(\mathbf{x}^*)$, while the upper bound (UB) is estimated by:

$$f\big(\mathbf{x}^*\big)+\sum_{\xi_s\in\Omega}\pi_s\big[\mathcal{Q}\big(\mathbf{x}^*,\xi_s\big)\big], \tag{58}$$

where

$$\begin{aligned}\mathcal{Q}\big(\mathbf{x}^*,\xi_s\big) = \min_{\mathbf{y}_s\in\mathcal{Y}(\mathbf{x}^*,\xi_s)}\ & g(\mathbf{y}_s)\\ \text{s.t. Eqns. } & (39)-(53)\\ & \mathbf{y}_s\in\mathbb{B}^{n_2}\times\mathbb{R}^m\end{aligned} \tag{59}$$

Based on [55, Th. 4], the algorithm now can converge in finite iterations. The detailed solution procedure is given

Algorithm 2: Two-Sample Average Approximation With Improved Benders Decomposition Algorithm

Data: $\mathcal{X}$,$\mathcal{Y}(\mathbf{x}, \xi_s)$, ξ_s, $\forall s$, and $l_{\max}$
Result: $\mathbf{x}$, $\mathbf{y}_s$, $\forall s$
Set $N = 10$, $\Delta N = 10$, $\alpha = 0.05$
while $N \leq N_{\max}$ **do**
  Randomly sample $2N$ samples using Monte-Carlo simulation, and $\pi_s = 1/N$
  For the first N samples, set $LB^0 = -\infty$, $UB^0 = \infty$, $l = 1$, **while** $l \leq l_{max}$ **do**
    Derive an optimal solution $\mathbf{x}^l$ by solving (57)
    if *Problem (57) is infeasible* **then**
      Terminate
    else
      Update $LB^l = \max\{LB^{l-1}, \text{objective value of MP}\}$
      for $s \in \Omega$
        Derive λ_s by solving (54)
        Derive $\mathcal{Q}'(\tilde{\mathbf{x}}, \xi_s)$ by solving (55)
        Derive cuts by solving (56)
      end
      Update $UB^l = \min\{UB^{l-1}, f(\mathbf{x}^l) + \sum_{\xi_s \in \Omega} \pi_s[\mathcal{Q}(\mathbf{x}^l, \xi_s)]\}$ by solving (59)
      if $(UB^l - LB^l)/\max(UB^l, LB^l) \leq \epsilon$ **then**
        Return $\mathbf{x}^l$, $\mathbf{y}_s^l$ and terminate
      else
        Add cuts (56) to the master problem (57)
        Update $l = l + 1$
      end
    end
  end
  For the rest N samples, calculate $\mathcal{Q}(\mathbf{x}^l, \xi_{N+1}), \ldots, \mathcal{Q}(\mathbf{x}^l, \xi_{2N})$
  Derive the upper bound of the confidence region using the Student's T-test:

$$\hat{UB} = \hat{\mu}_N + \hat{\sigma}_N \mathbf{Q}_{T_N(\alpha)/\sqrt{N}}$$

  where $\hat{\mu}_N$ and $\hat{\sigma}_N$ are the mean and standard deviation of $\mathcal{Q}(\mathbf{x}^l, \xi_{N+1}), \ldots, \mathcal{Q}(\mathbf{x}^l, \xi_{2N})$, and $\mathbf{Q}_{T_N}(\alpha)$ is the $(1-\alpha)$th quantile of Student's T-distribution with $N-1$ degrees of freedom
  if $UB^l \leq f(\mathbf{x}^l) + \hat{UB}$ **then**
    Return $\mathbf{x}^l$, $\mathbf{y}_s^l$ and terminate
  else
    $N = N + \Delta N$
  end
end

in Algorithm 2. Notably, the parameter N in problem (57) plays a critical role in balancing computational cost and the confidence level of the derived first-stage solution $\mathbf{x}$. If N is too small, the variation across Ω might be insufficient, leading to a lower confidence level of the derived solution $\mathbf{x}^l$. In this work, the two-sample average approximation with Monte-Carlo simulation and Student's T-distribution is employed to generate samples within Ω and test the confidence level of the derived solution [56].

2) Acceleration Techniques: The bigM reformulation of the NN model increases the complexity of the MILP problem significantly. We have already obtained relatively tight bounds as the M values and implemented weight sparsity and ReLU stability to simplify the reformulated NN constraints. However, even with these optimizations, solving the proposed UC problem with multiple integrated NNs remains challenging. To further accelerate the solution process, we have adopted the following techniques:

- Warm Start: The UC problem can be efficiently solved (within less than 1 second) when using simple proportional conversion loss models, despite their lower accuracy compared to the proposed NN surrogate models. Leveraging this advantage, the solutions obtained from the UC problem with proportional models integrated serve as excellent warm starts for the NN-integrated UC problem. This allows us to efficiently initialize the optimization process and improve convergence.
- Primal Cuts: We have incorporated the primal cuts technique introduced by [57] to further expedite convergence. The primal cuts involve including the constraints (39)-(53) of the subproblem with the largest objective value into the master problem in each iteration. By doing so, we can efficiently guide the optimization process towards feasible solutions, reducing the computational burden and enhancing convergence speed.

V. Case Studies

In this section, we begin by validating the impact of hard constraints and weight regularization on the performance of our proposed model. Additionally, we compare the approximation error of our model with that of analytical models. To further verify the effectiveness of our proposed model and solution method, we integrate the model with the proposed UC problem and conduct tests against other surrogate models. Lastly, we conduct hardware-in-loop experiments to assess the approximation accuracy of our proposed model.

A. Verification of Hard Constraints and Weight Regularization

1) NN Training: Four neural network models with the proposed structure, denoted as $\mathbf{u}_{\text{rec}}$, $\mathbf{u}_{\text{inv}}$, $\mathbf{u}_{\text{ch}}$ and $\mathbf{u}_{\text{dc}}$, are created to approximate the rectification, inversion, charging and discharging losses, respectively:

$$\mathbf{u}_{\text{rec}}(\mathbf{z}_{0,c}) \approx \mathcal{J}(\mathbf{z}_{0,c}) = p_{\text{L},c}^{\text{rec}} \tag{60}$$

$$\mathbf{u}_{\text{inv}}(\mathbf{z}_{0,c}) \approx \mathcal{J}(\mathbf{z}_{0,v}) = p_{\text{L},c}^{\text{inv}} \tag{61}$$

$$\mathbf{u}_{\text{ch}}(\mathbf{z}_{0,v}) \approx \mathcal{J}(\mathbf{z}_{0,c}) = p_{\text{L},v}^{\text{ch}} \tag{62}$$

$$\mathbf{u}_{\text{dc}}(\mathbf{z}_{0,v}) \approx \mathcal{J}(\mathbf{z}_{0,v}) = p_{\text{L},v}^{\text{dc}} \tag{63}$$

The range of the input power and voltage of $\mathbf{u}_c$ and $\mathbf{u}_v$ are set to be $\Omega_c = [0, 500] \times [658, 735]$ and $\Omega_v = [0, 200] \times [658, 735]$, respectively. The input data is normalized before be feeding into the neural networks:

TABLE III
NEURAL NETWORK PARAMETERS

Model	Layers	ϵ	λ_w	λ_{RS}	Fixed Sparsity	Pruning Threshold
AC/DC	[2, 30, 30, 30, 1]	1e-3	1e-8	1e-7	10%	1e-4
DC/DC	[2, 10, 10, 10, 1]	1e-3	1e-7	1e-7	10%	1e-4

[1] ϵ = training relative error threshold; λ_w = L1 regularization coefficient; λ_{RS} = ReLU stability regularization coefficient.

TABLE IV
TEST RESULTS OF APPROXIMATION ERRORS AND CALCULATION SPEED FOR DIFFERENT SURROGATE MODELS

Mode	Model	MAE (kW)	MRE (%)	Max RE (%)	Weight Sparsity (%)	Solving Time (s)
Rectification	Baseline	0.0062	0.18	203.27	0.00	67.63
	Constrained	0.0084	0.11	7.63	0.00	71.80
	Proposed	0.0082	**0.11**	**4.52**	62.06	**56.83**
Inversion	Baseline	0.0109	0.33	202.33	0.11	67.77
	Constrained	0.0207	0.26	8.07	0.00	71.15
	Proposed	0.0217	**0.24**	**4.86**	**41.27**	**50.78**
Charging	Baseline	0.0075	0.44	21.73	0.00	11.81
	Constrained	0.0047	0.19	5.05	0.00	12.00
	Proposed	**0.0044**	**0.18**	**5.44**	**21.74**	**10.98**
Discharging	Baseline	0.0074	0.44	24.36	0.00	11.53
	Constrained	0.0080	0.31	10.57	0.00	13.36
	Proposed	**0.0016**	**0.06**	**2.89**	**33.48**	**10.03**

[1] MAE = Mean Absolute Error; MRE = Mean Relative Error; Max RE = Maximum Relative Error.

$$\mathbf{z}_{0,c} = \left[\frac{p_c}{500}, \frac{V^{\text{DC}} - 658}{77}\right]^T \tag{64}$$

$$\mathbf{z}_{0,v} = \left[\frac{p_v}{200}, \frac{V^{\text{DC}} - 658}{77}\right]^T \tag{65}$$

According to [58], ReLU networks can approximate any continuous function with arbitrarily small errors using only three hidden layers. For each NN model, we fix the number of layers at 3 and tune the number of neurons in each layer based on the baseline model performance on a small subset of the training data. The neural network structures and other training hyperparameters are provided in Table III. All the neural networks are implemented and trained using Tensorflow [59] on a machine equipped with an NVIDIA GeForce RTX 3090 GPU.

2) Test Results of Trained NN Models: To verify the effectiveness of the hard constraint, weight sparsity and ReLU stability regularizations, we conducted comparisons on a test dataset of size 10,000 (solving times were evaluated on a subset of size 2,000 due to the complexity of the corresponding MILP problem) using the following three models, each with increasing levels of regularization:

1) Baseline Model: A basic fully connected model with no hard constraints or other additional regularizations.
2) Constrained Model: Baseline Model + Hard Constraints.
3) Proposed Model: Constrained Model + L1 regularization + Weight Pruning + Relu Stability.

The solving time of the NN surrogate itself is evaluated on the following zero objective problem for given p and V^{DC} values from the subset of the test dataset:

$$\min \ 0 \tag{66}$$

$$\text{s.t. } (4), (6), (19) - (24),$$

$$\mathbf{z}_0 = \left[p, V^{\text{DC}}\right]^T \tag{67}$$

The evaluation results are shown in Table IV. The introduction of hard constraints in the training of the Constrained Model effectively reduced relative errors while maintaining low absolute errors. However, the proposed model outperforms the Constrained Model in terms of relative errors, suggesting that the Constrained Model may suffer from overfitting without proper weight regularizations. Notably, constraint (11) is easier to satisfy than constraint (12), and the test results produced by the proposed model fully adhere to (11) (i.e., none of the test data violates constraint (11)). With the inclusion of L1 regularization and weight pruning, the weights of the proposed models exhibit high sparsity, leading to significant reductions in solving time.

B. Approximation Error Comparison of NN With Analytical Models

In this subsection, the performance of the proposed NN model is compared with two commonly used analytical models

TABLE V
COMPARISON OF APPROXIMATION ERROR OF THE PROPOSED MODEL WITH ANALYTICAL MODELS

Mode	Model	MAE (kW)	MRE (%)	Max RE (%)
Rectification	Proportional	0.9795	15.61	92.12
	Univariate	0.3007	6.18	2546.44
	Proposed	**0.0082**	**0.11**	**4.52**
Inversion	Proportional	0.9610	15.61	90.11
	Univariate	0.2994	6.12	1756.17
	Proposed	**0.0217**	**0.24**	**4.86**
Charging	Proportional	0.3560	17.02	96.76
	Univariate	0.0751	4.61	115.51
	Proposed	**0.0044**	**0.18**	**5.44**
Discharging	Proportional	0.3566	17.03	96.80
	Univariate	0.0759	0.61	115.85
	Proposed	**0.0016**	**0.06**	**2.89**

for MILP. The following two analytical models are created as comparative cases based on the literature review in Section I:

1) Proportional Model: The conversion loss is assumed to be proportional to the input power with a fixed conversion efficiency. The conversion losses are calculated by:

$$p_{\mathrm{L}} = (1 - \eta)p \tag{68}$$

where η is the fixed conversion efficiency. In this study the fixed conversion efficiency for AC/DC and DC/DC converters are set to be $\eta_c = 95.91\%$ and $\eta_v = 96.87\%$, respectively.

2) Univariate Model: The conversion loss is expressed as a univariate quadratic function of the current I:

$$p_{\mathrm{L}} = C_1 I^2 + C_2 I + C_3 \tag{69}$$

where C_1, C_2 and C_3 are the quadratic coefficients. The quadratic coefficients for AC/DC and DC/DC converters are set to be $\{C_{1,c} = -8.9077, C_{2,c} = 30.8996, C_{3,c} = 1.1816\}$ and $\{C_{1,v} = -8.9077, C_{2,v} = 30.8996, C_{3,v} = 1.1816\}$, respectively.

The conversion efficiencies and the quadratic coefficients are obtained by curve fitting on the same training dataset as the proposed model. The uncertain set Ω is obtained based on the uncertain modeling method in [10]. A visual representation of the uncertain variables can be found in Appendix A.

Table V shows the performance of different models on the test dataset. The NN model's adaptability and ability to approximate complex, nonlinear relationships dynamically enables it to outperform the analytical models with significantly lower MAE and MRE.

C. Influence of Variable DC Bus Voltage and Power Loss Surrogates

In this subsection, we conduct simulations of the proposed UC problem on a representative hybrid AC/DC MG system described in Section II. The aim is to investigate two key aspects:

(i) Impact of variable-voltage DC Bus control on power losses.
(ii) Computational efficiency of the proposed surrogates.

The detailed configurations of the MG system can be found in Appendix A. In total, we conducted simulations for seven distinct cases to analyze these aspects.

1) Case Setup:

- Cases 1 and 2: These cases are equipped with a fixed DC bus voltage and analytical conversion loss surrogates. They serve as benchmarks for comparing the impact of variable-voltage DC bus control on power losses (Aspect i).
- Cases 3-7: These cases focus on validating the computational efficiency of the proposed techniques (Aspect ii) for accelerating the UC solving process. Together, they form an ablation study, exploring the effectiveness of the following techniques outlined in previous sections:
  - Primal cuts (Section IV-D)
  - Warm start (Section IV-D)
  - Weight sparsity (WS) and ReLU stability (RS) regularization (Section III-E)

The detailed case setup, including the DC bus voltage settings, conversion loss surrogate models, the ablation study configurations are provided in Table VI. It is worth noting that without the primal cuts, the UC problem with the proposed surrogate model failed to converge to an optimal solution within the reasonable time limit (7200 seconds) using the basic Benders Decomposition algorithm. This highlights the necessity of developing acceleration techniques for NN-based MILP surrogates. Since cases omitting primal cuts did not converge in reasonable time, only one such case (Case 3) was included in the ablation study.

2) Simulation Results: The simulation results are presented in Table VI. Notably, cases with variable DC bus voltage exhibited lower total power losses compared to the fixed DC bus voltage cases, effectively demonstrating the efficacy of the multivariate loss formulation. Additionally, a comparison between case 4 and cases 5 and 6 revealed that both the Warm Start and WS+RS techniques significantly accelerated the solution speed. By combining the Warm Start and WS+RS techniques, the solution time was reduced by an impressive 79.62%.

It is essential to note that while the inclusion of primal cuts does increase the complexity of the master problem and extends the average solution time of iterations, it plays a pivotal role in reducing the overall number of iterations required for algorithm convergence. This optimization allows the proposed algorithm, as represented by Case 7, to converge in a mere 2 iterations, demonstrating its efficiency in solving the proposed problem within a reasonable timeframe.

Fig. 6 presents the total power losses for different scenarios in cases 1, 2 and 7. Notably, in Case 7, the power losses consistently show a reduction compared to cases 1 and 2, demonstrating the robustness of the loss minimization problem against disturbances from renewable energy sources and loads.

The optimization results of decision variables under different scenarios for Case 7 are illustrated in Fig. 7. Fig. 7(a) reveals that, for most of the time, the V^{DC} values are maintained at the lowest level (658 V) across all scenarios. The reduction of the DC bus voltage effectively decreases the total

TABLE VI
TOTAL POWER LOSSES AND SOLUTION TIME UNDER DIFFERENT CASES

Case #	DC Bus Voltage	Conversion Loss Surrogate	Speed Up Technique			Total Power Losses (kW)	Solution Time (s)	# of Iterations
			Primal Cuts	Warm Start	WS+RS			
Case 1	Fixed (700 V)	Proportional	✓			237.81	0.33	3
Case 2	Fixed (700 V)	Univariate	✓			237.03	99.19	3
Case 3	Variable	Proposed				/	Timeout	/
Case 4	Variable	Proposed	✓			223.77	4301.77	3
Case 5	Variable	Proposed	✓	✓		223.01	1881.51	2
Case 6	Variable	Proposed	✓		✓	223.28	1962.07	3
Case 7	Variable	Proposed	✓	✓	✓	223.15	**876.71**	2

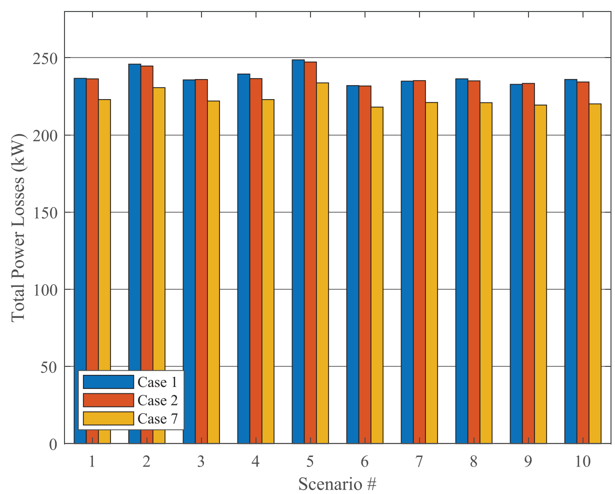

Fig. 6. Comparison of total power losses under different scenarios for case 1, 2, and 7.

conversion losses, as indicated by the conversion loss coefficient curve shown in Fig. 4. Furthermore, Fig. 7(b) illustrates that the bi-directional converters consistently operate in the rectifying mode, converting power from the AC side to the DC side, with no power converted in the opposite direction. This avoids additional conversion losses. Combining both Fig. 7(a) and 7(b), it becomes evident that when significant power conversion occurs, the DC bus voltage is controlled at its lowest level to minimize the conversion losses.

Regarding the ESS activity, Fig. 7(c) shows that during the period from 0:00 to 9:00, the ESS remains mostly idle due to the combined effect of low load demand and insufficient PV generation. At 10:00, the ESS is discharged to meet increased load demands. From 11:00 to 19:00, the ESS either charges or remains idle due to sufficient PV generation. After 20:00, as the PV generation decreases, the ESS returns to discharging mode. Overall, the ESS activity is kept to a minimum to reduce unnecessary conversion losses. These findings highlight the effectiveness of the optimization strategy in reducing power losses and optimizing the operation of the hybrid AC/DC MG under various scenarios.

D. Hardware-in-Loop Verification

The proposed model was verified using a hardware-in-loop (HIL) system shown in Fig. 8. The HIL system

TABLE VII
COMPARISON OF THE LOSSES GENERATED BY SURROGATE MODELS AND THE HIL SIMULATION RESULTS

Case #	Total Power Losses (kW)		MAE (kW)	MRE (%)
	Surrogate	HIL		
Case 1	237.81	236.90	0.1076	8.63
Case 2	237.03	236.69	0.0209	2.79
Case 7	223.15	224.01	**0.0098**	**0.93**

was constructed using the system-level simulation software PLECS [60], developed by Plexim GmbH. A simplified version of the hybrid AC/DC microgrid described in Section II was built in PLECS and simulated in real-time using the RT-BOX HIL system, as shown in Fig. 9. The digital signal processor (DSP) controller, based on the TMS320F28335 chip from Texas Instruments Incorporated (TI), was connected to the RT-BOX to control the currents in the system. The optimization results from the UC problem described in Section V-C were input into the HIL system to verify power losses under various scenarios. The HIL simulation process is detailed in Appendix B.

Table VII compares the total power losses generated by the HIL simulation with the surrogate-generated losses for cases 1, 2, and 7. The MAE and MRE were calculated across all types of losses (inversion, rectification, charging, and discharging losses). The proposed model (case 7) exhibits significantly smaller MAE and MRE compared to the analytical models (cases 1 and 2), showcasing superior accuracy against analytical models.

Fig. 11 further compares the HIL total losses for cases 1, 2, and 7, demonstrating that the HIL simulation, using the UC results generated by case 7, has the lowest total power losses under all scenarios. The proposed strategy (case 7) reduces power losses by 5.44% and 5.36% compared to fixed-voltage strategies (cases 1 and 2). This confirms the effectiveness of the variable DC bus voltage control in reducing power losses.

Fig. 10 presents a comparison between the HIL-simulated losses and the surrogate losses for case 7, revealing that the surrogate model slightly overestimates the losses. After further analysis of the errors, it is observed that the TMS320F28335 chip itself introduces inherent errors (±0.117 kW error

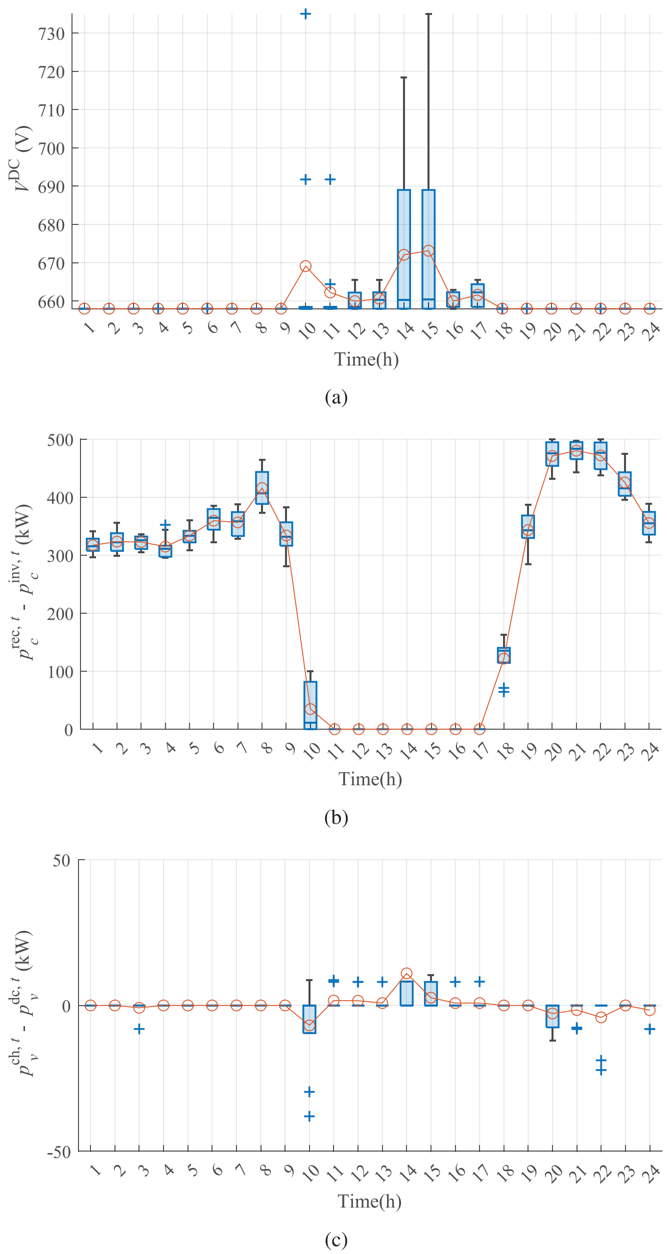

Fig. 7. Distribution (boxplots) and expected value (red line and circles) of decision variables under different scenarios for Case 7. (a) DC bus voltage (V^{DC}); (b) Power flow on the bi-directional AC/DC converter ($p_c^{\mathrm{rec},t} - p_c^{\mathrm{inv},t}$); (c) Power flow on the DC/DC converter ($p_{\mathrm{ESS},v}^{\mathrm{ch},t} - p_{\mathrm{ESS},v}^{\mathrm{dc},t}$).

for charging/discharging and ±0.287 kW error for rectification/inversion under given power limits). In addition, the analog signal conditioning (ADC) circuits between the HIL and control chip also contribute to errors. Furthermore, the HIL system exhibits larger relative errors when the target power is below 5% of the maximum power limits. As a considerable amount of power input into the converters in the UC optimization results falls below 5% of the maximum power limits, this leads to increased discrepancies between the HIL simulation results and the losses generated by surrogate models. Despite these built-in inaccuracies of the HIL system, the proposed surrogate model achieves a respectable 0.93% relative error.

In conclusion, the HIL verification results validate the proposed model and energy management scheme, effectively reducing power loss in EMS compared to conventional surrogate models.

VI. Conclusion

The surrogate modeling of the conversion losses for efficient EM is addressed in this paper. NN-based linear surrogate models are proposed to accurately and efficiently approximate these conversion losses. The NNs are trained with the augmented Lagrangian method to enforce additional physical and statistical constraints on the conversion loss-related variables. The conversion losses are modeled based on both the power and the DC bus voltage, taking advantage of the variable-voltage DC bus control. To mitigate computational costs, we employ weight sparsity and ReLU stability regularizations during training. Incorporating the surrogate model into a two-stage stochastic UC problem as a mixed-integer recourse enables optimizing the DC voltage to reduce power losses. An improved Benders decomposition algorithm with acceleration techniques is developed to solve the UC problem efficiently.

Comparative analysis against analytical models reveals the superior accuracy of our proposed NN model in approximating losses. To evaluate the effectiveness of our proposed EM strategy, we compare it to two other EM strategies where the DC bus voltages are fixed. The results demonstrate that our strategy reduces total power losses by 6.16% and 5.86% compared to the fix-voltage strategies. This confirms the effectiveness of optimizing the DC bus voltage to reduce power losses. Additionally, our strategy consistently yields lower total power losses across various scenarios, demonstrating its robustness under uncertainties. An ablation study validates the effectiveness of the NN weight optimization and solution acceleration techniques, reducing solution time by 79.62%. The combination of these techniques significantly reduces the computational cost associated with the NN surrogate.

To verify the practical usability of the proposed surrogate model and EM strategy, we conduct a HIL simulation. The proposed surrogate model achieves 0.93% MRE according to the HIL simulation results. Moreover, the proposed EM strategy exhibits power loss reductions of 5.44% and 5.36% compared to the fixed-voltage strategies in the HIL simulation.

In summary, the constrained NN conversion loss surrogates coupled with tailored solution techniques enable accurate modeling, robust optimization, and efficient computation of EM problems for variable-voltage hybrid MGs.

Appendix

A. Data Availability

We have made the uncertain vectors, NN parameters of the proposed models, UC optimization results, HIL simulation results, and microgrid configurations publicly available online at [61]. A representative portion of the data is displayed in Fig. 12, which showcases the distributions of the uncertain vectors across various scenarios.

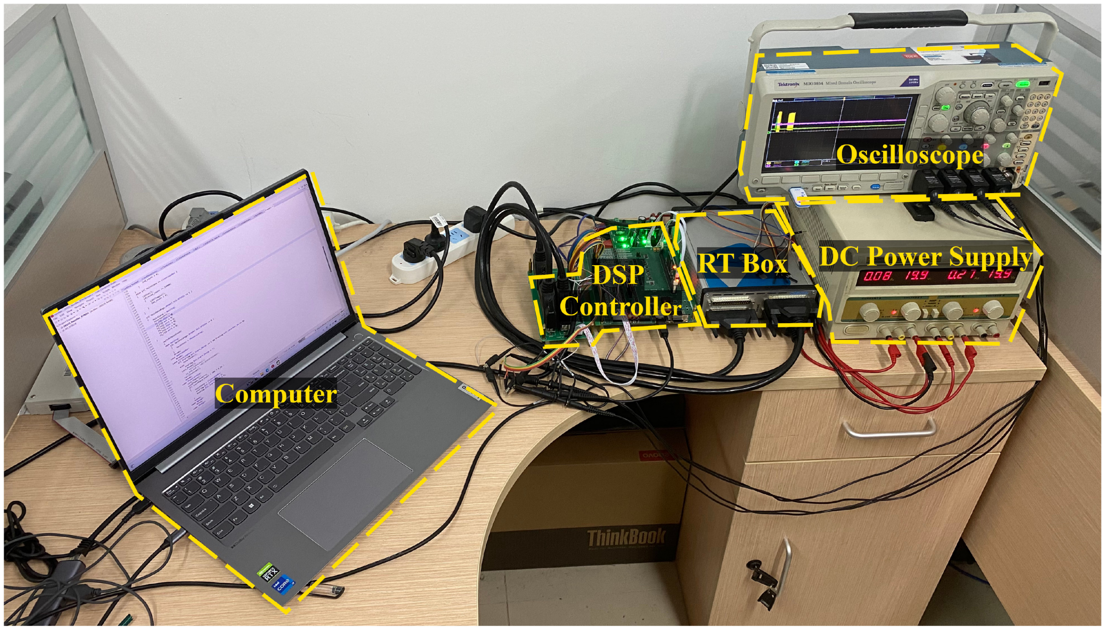

Fig. 8. Testing hardware of the hybrid AC/DC MG system.

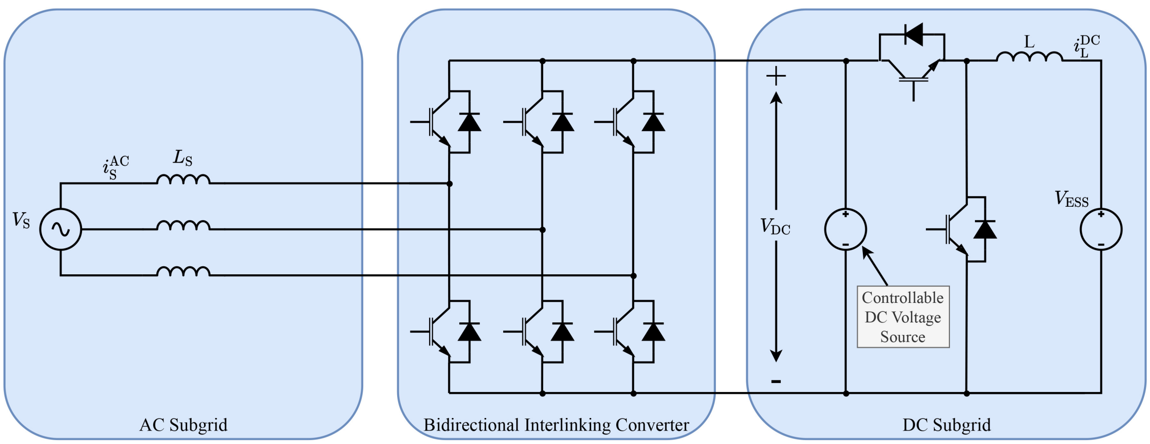

Fig. 9. Configuration of the system (a simplified version of the hybrid MG) simulated in the RT Box.

TABLE VIII
HIL System Parameter Values for AC and DC Subgrids

AC Subgrid			DC Subgrid		
Symbol	Description	Value	Symbol	Description	Value
V_{AC}	AC bus voltage	400 V	V_{DC}	DC bus voltage	[658, 735] V
f_{ref}	Rated frequency	50 Hz	V_{ESS}	ESS terminal voltage	400 V
L_S	Filter inductance	3 mH	L	Filter inductance	4 mH

B. HIL Simulation Process

As illustrated in Fig. 8, the HIL platform consists of an RT Box, a DSP controller, a DC power supply, an oscilloscope, and a computer. The real-time simulation of the simplified hybrid MG system, depicted in Fig. 9, is executed within the RT Box. During the simulation, the DSP controller generates control signals for the current loops. The computer is used for transmitting the circuit and simulation scripts to the DSP controller and the RT Box. Tables VIII and IX provide detailed parameter values for the AC and DC subgrids, as well as the converters, of the MG system. The IGBT modules of the converters are based on the Infineon FF1800R17IP5P IGBT module, with detailed parameters accessible on the producer's website.

The detailed HIL simulation process for a given case is as follows:

TABLE IX
HIL SYSTEM PARAMETER VALUES FOR AC/DC AND DC/DC CONVERTERS

AC/DC Converter			DC/DC Converter		
Symbol	Description	Value	Symbol	Description	Value
$P_{c,\max}$	Rated power	500 kW	$P_{v,\max}$	Rated power	200 kW
$f_{s,c}$	Switching frequency	10 kHz	$f_{s,v}$	Switching frequency	10 kHz

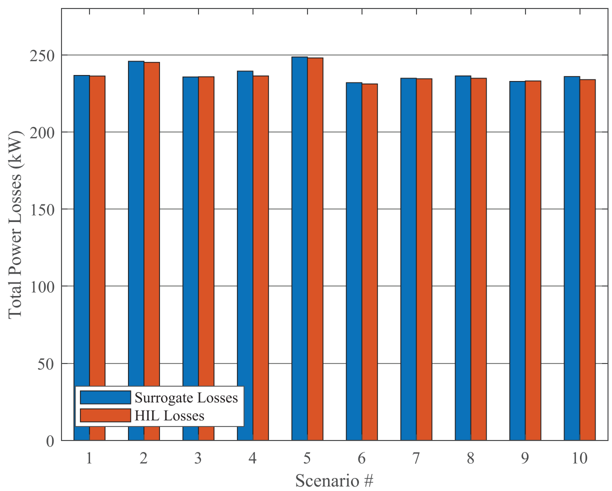

Fig. 10. Comparison of HIL simulated losses and the surrogate losses for Case 7.

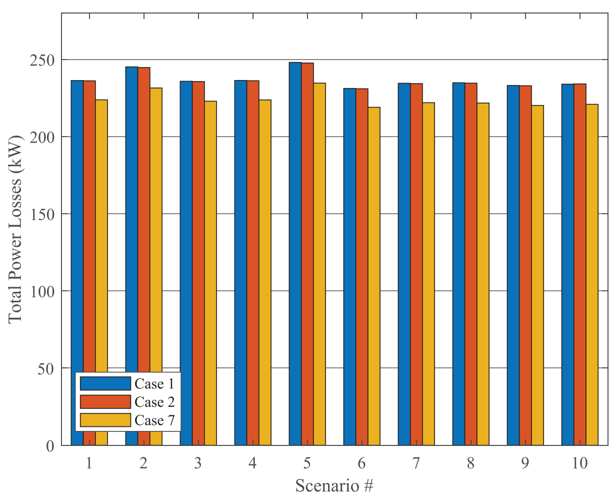

Fig. 11. Comparison of HIL simulated losses for different cases.

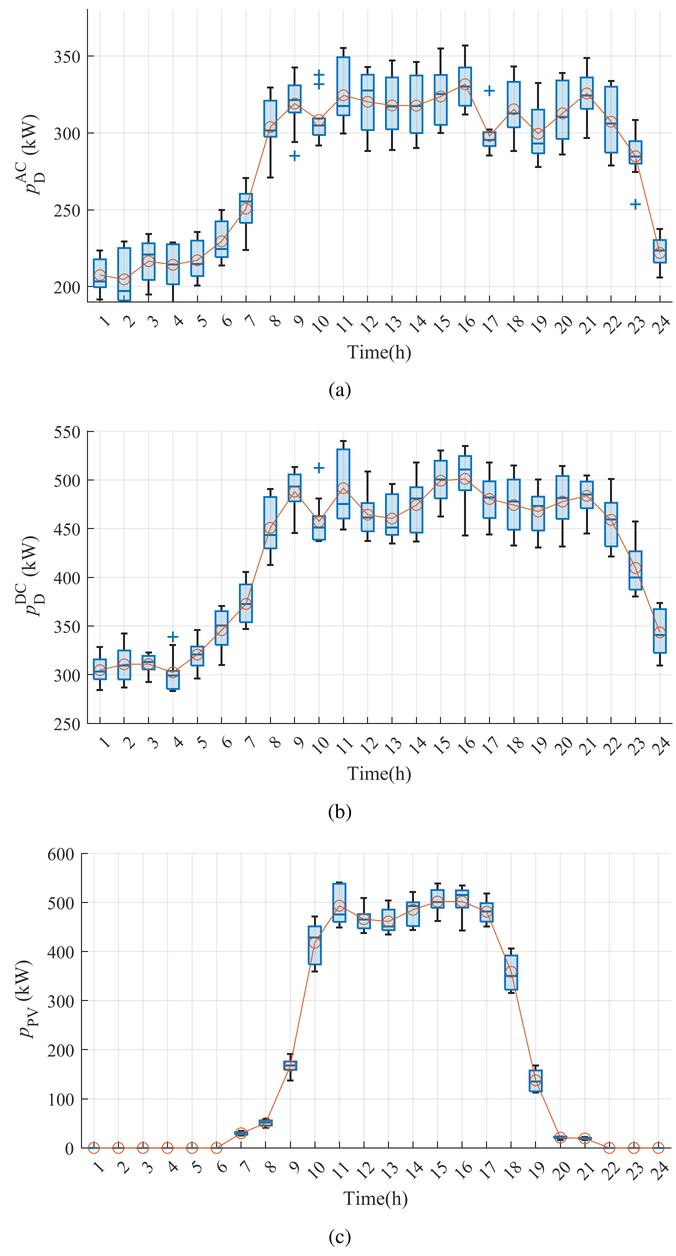

Fig. 12. Distribution (boxplots) and mean value (red line and circles) of the uncertain set under Case 7. (a) AC Load; (b) DC Load; (c) PV Generation.

1) Input Data Preparation: The optimization results of the UC problem, which include various power sources (p_g^t, p_{UG}^t), load sheddings ($p_{\mathrm{S}}^{\mathrm{AC},t}$, $p_{\mathrm{S}}^{\mathrm{DC},t}$), injected converter powers ($p_c^{\mathrm{rec},t}$, $p_c^{\mathrm{inv},t}$, $p_{\mathrm{ESS},v}^{\mathrm{ch},t}$, $p_{\mathrm{ESS},v}^{\mathrm{dc},t}$), and voltage magnitudes (V^{DC}), along with the uncertain power vectors ($p_{\mathrm{D}}^{\mathrm{AC},t}$, $p_{\mathrm{D}}^{\mathrm{DC},t}$, $\overline{P}_{\mathrm{PV}}^t$), are converted into AC and DC currents ($i_{\mathrm{S}}^{\mathrm{AC}}$, $i_{\mathrm{L}}^{\mathrm{DC}}$). Notably, for accurate conversion loss calculation in the HIL system, the output power of the converters is required. Therefore, $i_{\mathrm{S}}^{\mathrm{AC}}$ and $i_{\mathrm{L}}^{\mathrm{DC}}$ are calculated based on the output power of the converters, utilizing power balance equations (39) and (40). For each test case, a total of 10 scenarios $\times$ 24 timesteps $=$ 240 sets of ($i_{\mathrm{S}}^{\mathrm{AC}}$, $i_{\mathrm{L}}^{\mathrm{DC}}$, V^{DC}) values are generated and stored in C-Script files.
2) Real-Time Simulation on the RT-BOX: The C-Script files are transmitted to the RT-BOX by the computer

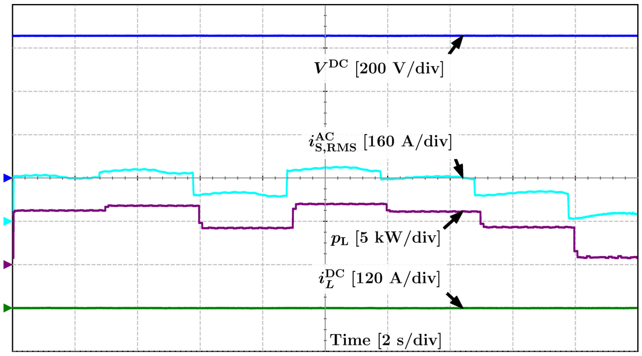

Fig. 13. Representative oscilloscope measurements captured during the HIL simulation of the proposed EM strategy in Case 7. The four signal channels are: (1) DC bus voltage V^{DC} (blue signal); (2) RMS current on the AC bus $i^{\mathrm{AC}}_{\mathrm{S,\,RMS}}$ (cyan signal); (3) average value of total power loss p_{L} (purple signal); (4) average DC inductor current $i^{\mathrm{DC}}_{\mathrm{L}}$ (green signal).

and executed in real-time, where the reference values of DC bus voltage and converter current are switched every 3 seconds ($t = 0, 3, 6, \ldots$). Each 3-second HIL simulation interval corresponds to one time interval (1 hour in our configuration) of MG operation. The DSP controller regulates the converters to follow the reference current ($i^{\mathrm{AC}}_{\mathrm{S}}, i^{\mathrm{DC}}_{\mathrm{L}}$), while the controllable DC voltage source adjusts the DC bus voltage. Due to the current loop control, the converters' current can track the reference value quickly. Meanwhile, the conversion losses are calculated in real time through C-Script in the RT-Box, utilizing the analytical model described in Section III-A.

3) Data Recording: The power losses quickly (in less than 2 seconds) reach a steady state after switching the reference values. Therefore, every two seconds ($t = 2, 5, 8, \ldots$) after following the reference value switches, the average power conversion losses are recorded and written to an external USB flash drive as steady state-values for the corresponding time interval. With a total of 240 sets of reference values for a given case, each taking 3 seconds to simulate, the overall simulation time for one case is 720 seconds.
4) Data Analysis: After simulating all 240 sets of data, the results stored in the external USB flash drive are retrieved and analyzed for further power loss verification.

While the verification data is collected and stored using the C-Script of the RT-BOX, the HIL simulation system also provides real-time monitoring and visualization capabilities through oscilloscopes, as illustrated in Fig. 13. This representative oscilloscope screen demonstrates several key aspects of the HIL simulation:

- For most of the duration, V^{DC} remains stable at a relatively low level (as shown in Fig. 7(a)) to minimize power losses. Given the absence of charging/discharging activities (as observed in Fig. 7(c)) in the optimization results, $i^{\mathrm{DC}}_{\mathrm{L}}$ (the green signal) also remains at 0 for the majority of the time.
- Sudden jumps of signals occur every 3 seconds, coinciding with changes in reference values. The p_{L} signal exhibits a 0.1-second delay compared to other signal changes since the oscilloscope displays the average value of total power loss over a 0.1-second period.
- While V^{DC} instantly adapts to the new reference value, the currents are controlled by the Digital Signal Processor (DSP) controller, which leads to some initial fluctuations at the beginning of each 3-second interval.
- Real-time calculations of total losses within the HIL system are based on the analytical model detailed in Section III-A. These calculations show the influence of both V^{DC} and currents on conversion losses, which can also be observed from the oscilloscope measurements.

References

[1] B. Lasseter, "Microgrids [distributed power generation]," in *Proc. IEEE Power Eng. Soc. Winter Meet.*, Columbus, OH, USA, 2001, pp. 146–149.

[2] R. H. Lasseter, "MicroGrids," in *Proc. IEEE Power Eng. Soc. Winter Meet.*, New York, NY, USA, 2002, pp. 305–308.

[3] M. Farrokhabadi et al., "Microgrid stability definitions, analysis, and examples," *IEEE Trans. Power Syst.*, vol. 35, no. 1, pp. 13–29, Jan. 2020.

[4] G. Zhang et al., "Forming a reliable hybrid microgrid using electric spring coupled with non-sensitive loads and ESS," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 2867–2879, Jul. 2020.

[5] M. Jasinski and S. Stynski, "AC–DC–AC converters for distributed power generation systems," in *Power Electronics for Renewable Energy Systems, Transportation and Industrial Applications*, Sebastopol, CA, USA: o'reilly, 2014, pp. 319–364.

[6] F. Nejabatkhah and Y. W. Li, "Overview of power management strategies of hybrid AC/DC microgrid," *IEEE Trans. power Electron.*, vol. 30, no. 12, pp. 7072–7089, Dec. 2015.

[7] J. Wang, C. Jin, and P. Wang, "A uniform control strategy for the interlinking converter in hierarchical controlled hybrid AC/DC microgrids," *IEEE Trans. Ind. Electron.*, vol. 65, no. 8, pp. 6188–6197, Aug. 2018.

[8] R. R. Kolluri, I. Mareels, and J. Hoog, "Controlling DC microgrids in communities, buildings and data centers," *IET Smart Grid*, vol. 3, no. 3, pp. 376–384, Jun. 2020.

[9] J. W. Kolar, F. Krismer, Y. Lobsiger, J. Muhlethaler, T. Nussbaumer, and J. Minibock, "Extreme efficiency power electronics," in *Proc. 7th Int. Conf. Integr. Power Electron. Syst. (CIPS)*. Nuremberg, Germany, 2012, pp. 1–22.

[10] Q. Xu, T. Zhao, Y. Xu, Z. Xu, P. Wang, and F. Blaabjerg, "A distributed and robust energy management system for networked hybrid AC/DC microgrids," *IEEE Trans. Smart Grid*, vol. 11, no. 4, pp. 3496–3508, Jul. 2020.

[11] S. Singh, P. Chauhan, M. A. Aftab, I. Ali, S. M. S. Singh, and T. S. Ustun, "Cost optimization of a stand-alone hybrid energy system with fuel cell and PV," *Energies*, vol. 13, no. 5, p. 1295, Mar. 2020.

[12] Z. Liang, H. Chen, X. Wang, S. Chen, and C. Zhang, "Risk-based uncertainty set optimization method for energy management of hybrid AC/DC microgrids with uncertain renewable generation," *IEEE Trans. Smart Grid*, vol. 11, no. 2, pp. 1526–1542, Mar. 2020.

[13] Q. Zhao, J. García-González, O. Gomis-Bellmunt, E. Prieto-Araujo, and F. M. Echavarren, "Impact of converter losses on the optimal power flow solution of hybrid networks based on VSC-MTDC," *Elect. Power Syst. Res.*, vol. 151, pp. 395–403, Oct. 2017.

[14] H. Siraj, N. U. Hassan, H. A. Khan, and C. Yuen, "Joint optimization of AC/DC conversion loss and battery lifetime in intermittent power systems," in *Proc. IEEE Innov. Smart Grid Technol.*, Singapore, 2018, pp. 446–451.

[15] B. Wei, X. Han, P. Wang, H. Yu, W. Li, and L. Guo, "Temporally coordinated energy management for AC/DC hybrid microgrid considering dynamic conversion efficiency of bidirectional AC/DC converter," *IEEE Access*, vol. 8, pp. 70878–70889, 2020.

[16] J.-S. Lai, L. Leslie, J. Ferrell, and T. Nergaard, "Characterization of HV-IGBT for high-power inverter applications," in *Proc. 14th IAS Annu. Meet. Conf. Rec Ind. Appl. Conf.*, Hong Kong, China, 2005, pp. 377–382.

[17] N. Eghtedarpour and E. Farjah, "Power control and management in a hybrid AC/DC microgrid," *IEEE Trans. Smart Grid*, vol. 5, no. 3, pp. 1494–1505, May 2014.

[18] P. Li, P. Han, S. He, and X. Wang, "Double-uncertainty optimal operation of hybrid AC/DC microgrids with high proportion of intermittent energy sources," *J. Modern Power Syst. Clean Energy*, vol. 5, no. 6, pp. 838–849, Nov. 2017.

[19] Z. Zhu, D. Liu, Q. Liao, F. Tang, J. J. Zhang, and H. Jiang, "Optimal power scheduling for a medium voltage AC/DC hybrid distribution network," *Sustainability*, vol. 10, no. 2, p. 318, Jan. 2018.

[20] B. Papari, C. S. Edrington, I. Bhattacharya, and G. Radman, "Effective energy management of hybrid AC–DC microgrids with storage devices," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 193–203, Jan. 2019.

[21] H. Qiu, W. Gu, Y. Xu, and B. Zhao, "Multi-time-scale rolling optimal dispatch for AC/DC hybrid microgrids with day-ahead distributionally robust scheduling," *IEEE Trans. Sustain. Energy*, vol. 10, no. 4, pp. 1653–1663, Oct. 2019.

[22] X. Wu, W. Cao, D. Wang, and M. Ding, "A multi-objective optimization dispatch method for microgrid energy management considering the power loss of converters," *Energies*, vol. 12, no. 11, p. 2160, Jun. 2019.

[23] E. Aprilia, K. Meng, H. H. Zeineldin, M. Al Hosani, and Z. Y. Dong, "Modeling of distributed generators and converters control for power flow analysis of networked islanded hybrid microgrids," *Elect. Power Syst. Res.*, vol. 184, Jul. 2020, Art. no. 106343.

[24] Y. Zhang, X. Meng, A. M. Shotorbani, and L. Wang, "Minimization of AC-DC grid transmission loss and DC voltage deviation using adaptive droop control and improved AC-DC power flow algorithm," *IEEE Trans. Power Syst.*, vol. 36, no. 1, pp. 744–756, Jan. 2021.

[25] D. M. Rao Korada and M. K. Mishra, "DC bus voltage control in hybrid AC/DC microgrid system," in *Proc. IEEE Int. Conf. Environ. Elect. Eng. IEEE Ind. Commer. Power Syst. Eur. (EEEIC/I CPS Europe)*. Madrid, Spain, 2020, pp. 1–6.

[26] X. Liu, M. A. Zagrodnik, and A. K. Gupta, "Electrical power systems," U.S. Patent 17 451 963, Aug. 2022.

[27] G. S. Thirunavukkarasu, M. Seyedmahmoudian, E. Jamei, B. Horan, S. Mekhilef, and A. Stojcevski, "Role of optimization techniques in microgrid energy management systems—A review," *Energy Strat. Rev.*, vol. 43, Sep. 2022, Art. no. 100899.

[28] K. Hornik, "Approximation capabilities of multilayer feedforward networks," *Neural Netw.*, vol. 4, no. 2, pp. 251–257, 1991.

[29] B. Aslan, S. Balci, A. Kayabasi, and B. Yildiz, "The core loss estimation of a single phase inverter transformer by using adaptive neuro-fuzzy inference system," *Measurement*, vol. 179, Jul. 2021, Art. no. 109427.

[30] K. Sabanci, S. Balci, and M. F. Aslan, "Estimation of the switching losses in DC-DC boost converters by various machine learning methods," *J. Energy Syst.*, vol. 4, no. 1, pp. 1–11, 2020.

[31] S. Devi, S. Makkapati, and R. Seyezhai, "Estimation of switching loss for boost DC –DC converter using neural networks," *Agr. Mech. Asia*, vol. 52, no. 1, p. 11, 2021.

[32] S. Kalker, D. Meier, C. H. van der Broeck, and R. W. De Doncker, "Self-calibrating loss models for real-time monitoring of power modules based on artificial neural networks," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*. Detroit, MI, USA, 2022, pp. 1–8.

[33] G. S. Misyris, A. Venzke, and S. Chatzivasileiadis, "Physics-informed neural networks for power systems," in *Proc. IEEE Power Energy Soc. General Meet.*, 2020, pp. 1–5.

[34] S. Zhao, Y. Peng, Y. Zhang, and H. Wang, "Parameter estimation of power electronic converters with physics-informed machine learning," *IEEE Trans. Power Electron.*, vol. 37, no. 10, pp. 11567–11578, Oct. 2022.

[35] H. Li, S. R. Lee, M. Luo, C. R. Sullivan, Y. Chen, and M. Chen, "MagNet: A machine learning framework for magnetic core loss modeling," in *Proc. IEEE 21st Workshop Control Model. Power Electron. (COMPEL)*. Aalborg, Denmark, 2020, pp. 1–8.

[36] H. Li et al., "How MagNet: Machine learning framework for modeling power magnetic material characteristics," *IEEE Trans. Power Electron.*, vol. 38, no. 12, pp. 15829–15853, Dec. 2023.

[37] H. Li, D. Serrano, S. Wang, T. Guillod, M. Luo, and M. Chen, "Predicting the B-H loops of power magnetics with transformer-based encoder-projector-decoder neural network architecture," in *Proc. IEEE Appl. Power Electron. Conf. Expo. (APEC)*. Orlando, FL, USA, 2023, pp. 1543–1550.

[38] H. R. Baghaee, M. Mirsalim, G. B. Gharehpetian, and H. A. Talebi, "Decentralized sliding mode control of WG/PV/FC microgrids under unbalanced and nonlinear load conditions for on- and off-grid modes," *IEEE Syst. J.*, vol. 12, no. 4, pp. 3108–3119, Dec. 2018.

[39] P. Chaudhary and M. Rizwan, "Energy management supporting high penetration of solar photovoltaic generation for smart grid using solar forecasts and pumped hydro storage system," *Renew. Energy*, vol. 118, pp. 928–946, Apr. 2018.

[40] M. E. Gamez Urias, E. N. Sanchez, and L. J. Ricalde, "Electrical microgrid optimization via a new recurrent neural network," *IEEE Syst. J.*, vol. 9, no. 3, pp. 945–953, Sep. 2015.

[41] E. Kuznetsova, Y.-F. Li, C. Ruiz, E. Zio, G. Ault, and K. Bell, "Reinforcement learning for microgrid energy management," *Energy*, vol. 59, pp. 133–146, Sep. 2013.

[42] E. Foruzan, L.-K. Soh, and S. Asgarpoor, "Reinforcement learning approach for optimal distributed energy management in a microgrid," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 5749–5758, Sep. 2018.

[43] Y. Du and F. Li, "Intelligent multi-microgrid energy management based on deep neural network and model-free reinforcement learning," *IEEE Trans. Smart Grid*, vol. 11, no. 2, pp. 1066–1076, Mar. 2020.

[44] C. Wang, Q. Duan, W. Gong, A. Ye, Z. Di, and C. Miao, "An evaluation of adaptive surrogate modeling based optimization with two benchmark problems," *Environ. Model. Softw.*, vol. 60, pp. 167–179, Oct. 2014.

[45] T. Zhao, X. Liu, P. Wang, and F. Blaabjerg, "More efficient energy management for networked hybrid AC/DC microgrids with multivariable nonlinear conversion losses," *IEEE Syst. J.*, vol. 17, no. 2, pp. 3212–3223, Jun. 2023.

[46] X. Liu, 2023, "Conversion loss data generation with analytical models," IEEE DataPort. [Online]. Available: https://dx.doi.org/10.21227/wnw5-hj68

[47] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, Jan. 1989.

[48] M. Toussaint, (Stanford Univ., Stanford, CA, USA). *Introduction to Optimization*. (2014). [Online]. Available: https://www.user.tu-berlin.de/mtoussai/teaching/14-Optimization/14-Optimization- script.pdf

[49] L. Lu, R. Pestourie, W. Yao, Z. Wang, F. Verdugo, and S. G. Johnson, "Physics-informed neural networks with hard constraints for inverse design," *SIAM J. Sci. Comput.*, vol. 43, no. 6, pp. B1105–B1132, 2021.

[50] V. Tjeng, K. Y. Xiao, and R. Tedrake, "Evaluating robustness of neural networks with mixed integer programming," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–21. [Online]. Available: https://openreview.net/forum?id=HyGIdiRqtm

[51] R. C. Moore and J. DeNero, "L1 and L2 regularization for multiclass hinge loss models," in *Proc. Symp. Mach. Learn. Speech Lang. Process.*, 2011, pp. 1–5.

[52] M. Zhu and S. Gupta, "To prune, or not to prune: Exploring the efficacy of pruning for model compression," 2017, *arXiv:1710.01878*.

[53] K. Y. Xiao, V. Tjeng, N. M. Shafiullah, and A. Madry, "Training for faster adversarial robustness verification via inducing ReLU stability," 2019, *arXiv:1809.03008*.

[54] J. BnnoBRs, "Partitioning procedures for solving mixed-variables programming problems," *Numer. Math*, vol. 4, no. 1, pp. 238–252, 1962.

[55] J. Zou, S. Ahmed, and X. A. Sun, "Nested decomposition of multistage stochastic integer programs with binary state variables," *Optim. Online*, vol. 5436, p. 34, May 2016.

[56] D. Bertsimas, V. Gupta, and N. Kallus, "Robust sample average approximation," *Math. Program.*, vol. 171, pp. 217–282, Sep. 2018.

[57] X. Qi, T. Zhao, X. Liu, and P. Wang, "Three-stage stochastic unit commitment for microgrids towards frequency security via renewable energy deloading," *IEEE Trans. Smart Grid*, vol. 14, no. 6, pp. 4256–4267, Nov. 2023.

[58] M. Telgarsky, "Deep learning theory lecture notes," Lecture Notes v0.0-e7150f2d (alpha), Univ. Illinois Urbana-Champaign, Champaign, IL, USA, Oct. 2021. [Online]. Available: https://mjt.cs.illinois.edu/dlt/

[59] M. Abadi et al., *TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems*," (2015). TensorFlow. [Online]. Available: https://www.tensorflow.org/

[60] P. GmbH, *PLECS User Manual*, Plexim GmbH, Zurich, Switzerland, 2002.

[61] X. Liu, 2023, "More efficient energy management for hybrid AC/DC microgrids with hard constrained neural network-based conversion loss surrogate models," IEEE DataPort. [Online]. Available: https://dx.doi.org/10.21227/n4x4-yg82

Xuan Liu (Graduate Student Member, IEEE) received the B.Sc. degree in computer science from the University of San Francisco in 2015, and the M.Sc. degree in computer science from Washington State University in 2019. He is currently pursuing the Ph.D. degree in computer science with Jinan University, Guangdong, China. His research interests include explainable artificial intelligence, optimization, and microgrid energy management.

Yuanze Wang received the B.E. degree in automation from Yangzhou University, Jiangsu, China, in 2020, and the master's degree in electronic information from Jinan University, Guangdong, China, in 2023. His research interests include power electronics and motor drive control.

Xiong Liu (Senior Member, IEEE) received the B.E. and M.Sc. degrees in electrical engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2006 and 2008, respectively, and the Ph.D. degree in electrical engineering from the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, in 2013.

From July to November 2008, he was an Engineer with Shenzhen Nanrui Technologies Company Ltd., Shenzhen, China. From September 2011 to January 2012, he was a Visiting Scholar with the Department of Energy Technology, Aalborg University, Aalborg East, Denmark. From April 2012 to December 2013, he was a Researcher with the Energy Research Institute, Nanyang Technological University. From December 2013 to July 2020, he was a Principal Technologist with Rolls-Royce Electrical, Rolls-Royce Singapore Pte. Ltd., Singapore. He is currently an Associate Professor with the Energy Electricity Research Center, International Energy College, Jinan University, Zhuhai, China. His research interests include power electronics, motor drive, and electrical/hybrid propulsion system for marine and aerospace.

Dr. Liu was the recipient of the Best Paper Award at the IEEE International Power Electronics and Motion Control Conference-Energy Conversion Congress and Exposition Asia in 2012.

Tianyang Zhao (Senior Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in electrical engineering from North China Electric Power University, Beijing, China, in 2011, 2013, and 2017, respectively. He was a Postdoctoral Research Fellow with the Energy Research Institute, Nanyang Technological University, Singapore, from July 2017 to August 2020. He is currently an Associate Professor with Energy and Electricity Research Center, Jinan University. His research interests include operations research and resilience.

Peng Wang (Fellow, IEEE) received the B.Sc. degree in electrical engineering from Xi'an Jiaotong University, Xi'an, China, in 1978, the M.Sc. degree from the Taiyuan University of Technology, Taiyuan, China, in 1987, and the M.Sc. and Ph.D. degrees in electrical engineering from the University of Saskatchewan, Saskatoon, SK, Canada, in 1995 and 1998, respectively.

He is currently a Professor with Nanyang Technological University, Singapore. His current research interests include power system planning and operation, renewable energy planning, solar/electricity conversion systems, and power system reliability analysis.

Dr. Wang is an Associate Editor or a Guest Editor-in-Chief of IEEE TRANSACTIONS ON SMART GRID, IEEE TRANSACTIONS ON POWER DELIVERY, *Journal of Modern Power Systems and Clean Energy*, and *CSEE Journal of Power and Energy Systems*.